Exact Second-Order Asymptotics for the Gaussian Gray–Wyner Problem

Lin Zhou and Lei Yu

Abstract

We revisit the Gray-Wyner problem of lossy compression of bivariate Gaussian sources under the quadratic distortion measure and derive the exact second-order asymptotics. Specifically, we first identify the boundary rate tuples using the explicit first-order asymptotics by Yu (TIT 2023) and then characterize the tradeoff among three second-order coding rates subject to a constraint on the joint excess-distortion probability. Our main contribution is the achievability proof, where we derive a covariance shell covering lemma that is analogous to the type covering lemma for the discrete case. In particular, our covering lemma classifies the source sequences into power shell types via their empirical covariance matrix. For each such power shell type, we derive sufficient conditions on the compression rates so that covering within distortion levels are achievable. Finally, asymptotical analysis via the Berry-Esseen theorem yields the desired result.

Index Terms

Dispersion, Rate-distortion, Covering Lemma, Multiterminal source coding, Lossy source coding

I. Introduction

Data compression, a.k.a., source coding, was pioneered by Shannon [1] in this seminal paper. Specifically, Shannon proved that the entropy is the minimal compression rate for lossless compression of a discrete memoryless source (DMS) with vanishing error probability as the blocklength tends to infinity. A decade later, Strassen [2] refined Shannon's result by deriving the second-order asymptotics that approximates the finite blocklength performance of optimal codes by characterizing the order and the coefficient of the second dominant term of the non-asymptotic compression rate as a function of the blocklength and a given error probability. Recently, the result of Strassen was further refined by Kontoyiannis and Verdú who derived further refined third-order asymptotics [3]. As demonstrated in [2], [3], refined asymptotics approximates the performance of optimal codes at finite blocklengths.

Although insightful, lossless compression is valid only for a DMS. For continuous memoryless sources, lossy source coding, a.k.a., rate-distortion was introduced by Shannon [4]. In lossy source coding, the reconstruction is no longer required to be exactly the same as the original source sequence and the imperfection is measured via a distortion function. Shannon [4] derived the first-order asymptotics for lossy source coding under the expected distortion criterion when the blocklength tends to infinity and named the coefficient of the first dominant term of the minimal compression rate of optimal codes as the rate-distortion function. The same result also holds when one uses a vanishing excess-distortion probability criterion. Several decades later, the second-order asymptotics of lossy source coding was derived by Ingber and Kochman [5] and Kostina and Verdú [6]. Furthermore, for an equiprobable DMS under Hamming distortion, Kostina and Verdú also characterized the third-order asymptotics [6, Theorem 27].

Beyond point-to-point (P2P) data compression, motivated by practical applications in the storage/retrieval of correlated sources [7, Section V-C] and cache-aided delivery of correlated content [8], Gray and Wyner [9] proposed a source coding problem in which a pair of correlated source sequences is compressed, in either a lossless or lossy manner, and transmitted from a single transmitter over a common link and two private links to two different receivers. The first-order asymptotics, a.k.a., the rate region for the lossless case and the rate-distortion region for the lossy case, was characterized in [9]. Ever since, the problem is known as the Gray-Wyner problem, a classical multiterminal source coding problem concerning compression of correlated memoryless sources. Several decades later, for the DMS, the second-order asymptotics for the Gray-Wyner problem was derived by Watanabe [10] for the lossless case and by Zhou, Tan and Motani [11] for the lossy case. Furthermore, Liu et al. [12, Theorem 17] derived second-order converse bounds for both discrete correlated sources under bounded distortion measures and bivariate Gaussian sources under the quadratic distortion measure. Recently, for bivariate Gaussian sources, the explicit form of the rate-distortion region was completely characterized by Yu [13, Theorem 3 and Corollary 2]. Throughout the paper, we refer to the Gray-Wyner problem for lossy compression of bivariate Gaussian memoryless sources under the quadratic distortion measure as the Gaussian Gray-Wyner problem.

In contrast to the discrete case, the exact second-order asymptotics for the Gaussian Gray-Wyner problem remains open until now. The key challenge lies in how to accurately extract the common information from bivariate Gaussian sources and derive the Gaussian analogue of type covering lemma [11, Lemma 8]. The method of types [14], tailed and used extensively for the DMS, cannot be applied to continuous sources like Gaussian sources studied in this paper.

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A. Main Contributions

We address the above challenge and derive exact second-order asymptotics for the Gaussian Gray-Wyner problem. Our main contribution is an achievability bound that matches the existing converse bound of Liu et al. [12, Theorem 17]. In particular, we derive a covariance shell covering lemma that is parallel to the type covering lemma for the DMS. To do so, we first define Gaussian type classes by grouping pairs of correlated Gaussian source sequences according to their empirical covariance matrix, generalizing the power type class for a single Gaussian source in the successive refinement problem [15, Section VI-A]. Subsequently, for each Gaussian type class, we use random covering for the common message and use Euclidean ball covering for the private messages. In particular, we show that the achievable common rate equals the minimum common rate derived from the first-order asymptotics when the source distribution is given by the Gaussian type class and two private rates are set as desired values. Finally, we use the Berry–Esseen theorem [16, Theorem 1.1] to conclude the achievability analysis. For the converse part, we combine the second-order converse bounds for the Gaussian Gray–Wyner problem of Liu et al. [12, Theorem 17] and the P2P Gaussian lossy source coding problem of Kostina and Verdú [6, Theorem 40].

We remark that concurrent with our study, Cao [17, Theorem 18] provided a similar result with a proof sketch but no concrete proof steps. Compared with [17, Theorem 18], we provide an explicit analysis of the bivariate Gaussian Gray–Wyner problem at boundary rate tuples with positive common rate. Specifically, we classify the boundary rate tuple into four regimes, calculate the supporting weights and dispersion in closed form and develop a covariance-shell covering code with common and private codebook bounds .For the converse, we identify the entropy-support quantity and its fluctuation, explain the specialization of the converse result of [12, Theorem 17], and handle zero supporting weights through a scalar reduction to the P2P case using [6, Theorem 40].

B. Other Related Works

Besides the Gray–Wyner problem, the second-order asymptotics has also been established for other multiterminal source coding problems. Specifically, Tan and Kosut [18] studied the Slepian–Wolf problem [19] of lossless compression of correlated sources, which was generalized by Nomura and Han [20] to mixed sources. Furthermore, No, Ingber, and Weissman [21] studied the successive refinement problem [22] under separate excess-distortion probabilities criteria while by Zhou, Tan, and Motani [15] considered the joint excess-distortion probability criterion. For the Wyner-Ziv problem [23] of lossy source coding with decoder side information, achievability result was obtained by Watanabe, Kuzuoka, and Tan [24], which was recently refined by Li and Li [25]; very recently, exact second-order asymptotics was characterized by Zhou [26] under mild conditions. For a comprehensive summary of second-order asymptotics of lossy source coding, refer to the monograph [27].

The Gray–Wyner problem is also closely related to the study of common information. This line of research was pioneered by Wyner [28], who characterized the minimum common rate for the lossless Gray-Wyner problem when the sum rate of three encoders equals the joint entropy of the two correlated sources. Subsequently, Viswanatha, Akyol, and Rose [7] extended [28] to the lossy case and explicitly calculated the lossy common information. For the Gaussian Gray-Wyner problem, Sula and Gastpar [29] determined the minimum common rate subject to the sum of two private rates.

C. Organization for the Rest of the Paper

In Section II, we formulate the Gaussian Gray-Wyner problem, define the first- and second-order asymptotics and recall existing result. In Section III, we present and discuss our main results. The achievability and converse proofs of our main result are presented in Sections IV and V, respectively. Finally, Section VI concludes the paper and discusses future directions. For smooth presentation, the proofs of supporting lemmas are deferred to the appendices.

Notation

Upper-case letters denote random variables, and lower-case letters denote their realizations. Bold upper-case letters denote random vectors and matrices, while bold lower-case letters denote deterministic vectors and matrices. We use $(\mathbb{N}, \mathbb{R}, \mathbb{R}_+, \mathbb{R}_{+,0})$ to denote the sets of positive integers, real numbers, positive real numbers and non-negative real numbers, respectively. For any real number $x \in \mathbb{R}$, we define $[x]^+ := \max\{x, 0\}$. For any two integers $(a, b) \in \mathbb{N}^2$ such that $a \leq b$, we define $[a : b] := \{a, a+1, \ldots, b\}$ and use $[a]$ to denote $[1 : a]$ when $a \geq 1$. Given $d \in \mathbb{N}$, $\boldsymbol{\mu} \in \mathbb{R}^d$ and $\mathbf{\Sigma} \in \mathbb{S}_+^d$, we use $\mathcal{N}(\boldsymbol{\mu}, \mathbf{\Sigma})$ to denote the d-dimensional Gaussian distribution with mean vector $\boldsymbol{\mu}$ and covariance matrix $\mathbf{\Sigma}$. We use Φ to denote the cumulative distribution function (cdf) for the standard Gaussian random variable with mean zero and variance one and use $\mathrm{Q} := 1 - \Phi$ to denote its complementary cdf. We write $(\mathsf{E}_Q[\cdot], \mathsf{Var}_Q[\cdot], \mathsf{Cov}_Q(\cdot, \cdot))$ for expectation, variance, and covariance with respect to a distribution Q. We use the standard asymptotic notation such as $(O(\cdot), o(\cdot))$. All logarithms are natural. We use the scalar symbols (R_j, L_j, D_j, M_j, V) for rates, second-order coding rates, distortions, alphabet sizes, and dispersion, respectively.

Fix an integer $d \in \mathbb{N}$. We use $(\mathbb{S}^d, \mathbb{S}_+^d, \mathbb{S}_{++}^d)$ to denote the real symmetric, positive-semidefinite, and positive-definite $d \times d$ matrices. Fix any two matrices $(\mathbf{a}, \mathbf{b}) \in (\mathbb{S}^d)^2$, we write $\mathbf{a} \preceq \mathbf{b}$ when $\mathbf{b} - \mathbf{a} \in \mathbb{S}_+^d$, use $|\mathbf{a}|$ to denote the determinant, $\mathrm{tr}(\mathbf{a})$ to denote the trace, $\mathbf{a}^\mathsf{T}$ to denote the transpose, $\|\mathbf{a}\| = \sqrt{\langle \mathbf{a}, \mathbf{a} \rangle}$ to denote the Frobenius norm, and define $\langle \mathbf{a}, \mathbf{b} \rangle := \mathrm{tr}(\mathbf{a}^\mathsf{T} \mathbf{b})$. For

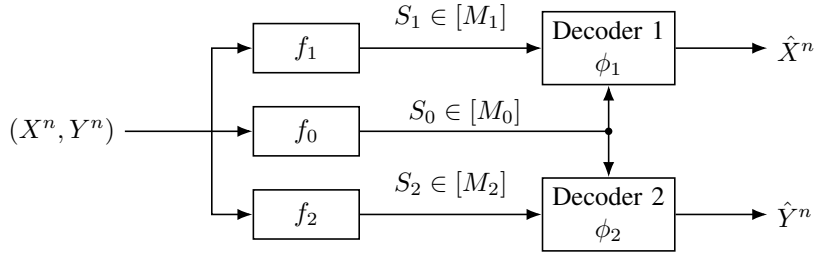


Fig. 1. System model for the Gaussian Gray–Wyner problem.

each $j \in [d]$, we use $\mathbf{e}_j \in \mathbb{R}^d$ to denote the d-dimensional vector with one being the j-th element and zero elsewhere. For a vector $\boldsymbol{\mu} \in \mathbb{R}^d$ and a real number $r \in \mathbb{R}_+$, we use $\mathcal{B}_d(\boldsymbol{\mu}, r) := \{\mathbf{x} \in \mathbb{R}^d : \|\mathbf{x} - \boldsymbol{\mu}\| \le r\}$ to denote the d-dimensional ball with center $\boldsymbol{\mu}$ and radius r; with zero radius, $\mathcal{B}_d(\boldsymbol{\mu}, 0) := \{\boldsymbol{\mu}\}$. For each $j \in [d]$, given a vector $\mathbf{v} \in \mathbb{R}^d$, we use $[\mathbf{v}]_j$ to denote its j-th coordinate; given another integer $n \in \mathbb{N}$ and a matrix $\mathbf{x}^n$ of dimension $d \times n$, we use $(\mathbf{x}^n)_j$ to denote its j-th row. We write $\mathbf{0}_d := (0, \ldots, 0)^\mathsf{T} \in \mathbb{R}^d$ and $\mathbf{0}_{d\times d} \in \mathbb{R}^{d\times d}$ for the all zero vector and all zero matrix, respectively, and use $\mathbf{1}_d$ for the $d \times d$ identity matrix.

II. Problem Formulation and Existing Results

A. System Model and Definitions

Fix positive real numbers $(\sigma_X, \sigma_Y, \rho_{XY}) \in \mathbb{R}_+^2 \times (0, 1)$, two distortion levels $(D_1, D_2) \in (0, \sigma_X^2) \times (0, \sigma_Y^2)$ and four positive integers $(n, M_0, M_1, M_2) \in \mathbb{N}^3$. Define the joint source distribution P_{XY}:

$$P_{XY} := \mathcal{N}(\mathbf{0}_2, \mathbf{\Sigma}_{XY}),\ \mathbf{\Sigma}_{XY} := \begin{bmatrix} \sigma_X^2 & \rho_{XY}\sigma_X\sigma_Y \\ \rho_{XY}\sigma_X\sigma_Y & \sigma_Y^2 \end{bmatrix} \in \mathbb{S}_{++}^2. \tag{1}$$

Let (P_X, P_Y) be the induced marginal distributions. We consider memoryless sources $(X^n, Y^n) \in (\mathbb{R}^n)^2$ that is generated i.i.d. from P_{XY}. As shown in Fig. 1, in the Gaussian Gray-Wyner problem, three encoders compress a pair of correlated sequences (X^n, Y^n) via three encoders (f_0, f_1, f_2), which are used by two decoders (ϕ_1, ϕ_2) to generate source sequence estimates $(\hat{X}^n, \hat{Y}^n)$, respectively. In particular, encoder f_0 sends the common message S_0 to both decoders while encoder f_i, for each $i \in [2]$, sends a private message S_i to decoder ϕ_i. To evaluate the reconstruction quality, we use the quadratic distortion measure. Specifically, given any $(x, \hat{x}) \in \mathbb{R}^2$ and $(x^n, \hat{x}^n) \in (\mathbb{R}^n)^2$, the quadratic distortion measure and its block extension are defined as follows:

$$d(x, \hat{x}) := (x - \hat{x})^2, \tag{2}$$

$$d(x^n, \hat{x}^n) := \frac{1}{n}\sum_{i\in[n]} d(x_i, \hat{x}_i) = \frac{1}{n}\left\|x^n - \hat{x}^n\right\|^2. \tag{3}$$

We can now define a code that mathematically defines the behavior of the encoders and decoders.

Definition 1. *An (n, M_0, M_1, M_2)-code consists of three encoders f_i, $i \in [0:2]$, and two decoders ϕ_j, $j \in [2]$ such that*

$$f_i : (\mathbb{R}^n)^2 \longrightarrow [M_i], \tag{4}$$

$$\phi_j : [M_0] \times [M_j] \longrightarrow \mathbb{R}^n. \tag{5}$$

Given an (n, M_0, M_1, M_2)-code and a pair of sequences $\mathbf{x}^n = (x^n, y^n) \in \mathbb{R}^{2\times n}$, the reconstructed sequences are

$$\hat{x}^n := \phi_1(f_0(\mathbf{x}^n), f_1(\mathbf{x}^n)), \qquad \hat{y}^n := \phi_2(f_0(\mathbf{x}^n), f_2(\mathbf{x}^n)). \tag{6}$$

To define the performance criterion, define the following excess-distortion event:

$$\mathcal{E}_n := \{\mathbf{x}^n \in \mathbb{R}^{2\times n} : d(x^n, \hat{x}^n) > D_1,\ \text{or}\ d(y^n, \hat{y}^n) > D_2\},. \tag{7}$$

Thus, $\mathcal{E}_n^c = \mathbb{R}^{2\times n} \setminus \mathcal{E}_n$ is the non-excess-distortion event. Define the joint excess-distortion probability as

$$\varepsilon_n(D_1, D_2) := P_{XY}^n(\mathcal{E}_n) = 1 - P_{XY}^n(\mathcal{E}_n^c). \tag{8}$$

For ease of notation, we use ε_n to denote $\varepsilon_n(D_1, D_2)$. The first-order asymptotics, a.k.a., the rate-distortion region, is defined as follows.

Definition 2. *A rate tuple $(R_0, R_1, R_2) \in \mathbb{R}^3_{+,0}$ is said to be first-order achievable if there exists a sequence of (n, M_0, M_1, M_2) codes such that for each $j \in [0:2]$,*

$$\limsup_{n\to\infty} \frac{\log M_{j,n}}{n} \le R_j, \qquad \lim_{n\to\infty} \varepsilon_n = 0. \tag{9}$$

The convex closure of all first-order achievable rate tuples is the first-order rate region, denoted by $\mathcal{R}(D_1, D_2|P_{XY})$.

In existing studies including the original paper of Gray and Wyner [9], the performance criterion is the following expected distortion criteria instead of the vanishing excess-distortion probability:

$$\limsup_{n\to\infty} \mathsf{E}_{P^n_{XY}}[d(X^n, \hat{X}^n)] \le D_1, \tag{10}$$

$$\limsup_{n\to\infty} \mathsf{E}_{P^n_{XY}}[d(Y^n, \hat{Y}^n)] \le D_2. \tag{11}$$

Denote the corresponding first-order rate region under the expected distortion criterion as $\mathcal{R}_{\mathrm{E}}(D_1, D_2|P_{XY})$. The following lemma shows that the two distortion criteria lead to the same first-order rate region for bivariate Gaussian source.

Lemma 1. *It follows that*

$$\mathcal{R}(D_1, D_2|P_{XY}) = \mathcal{R}_{\mathrm{E}}(D_1, D_2|P_{XY}) = \mathcal{R}^*(D_1, D_2|P_{XY}), \tag{12}$$

where

$$\mathcal{R}^*(D_1, D_2|P_{XY}) := \bigcup_{\substack{P_{W|XY}, P_{\hat{X}|XW}, P_{\hat{Y}|YW}: \\ \mathsf{E}[(X-\hat{X})^2]\le D_1,\ \mathsf{E}[(Y-\hat{Y})^2]\le D_2}} \big\{(R_0, R_1, R_2) \in \mathbb{R}^3_{+,0} : \tag{13}$$

$$R_0 \ge I(X, Y; W),\ R_1 \ge I(X; \hat{X}|W),\ R_2 \ge I(Y; \hat{Y}|W)\big\}. \tag{14}$$

Proof. Gray and Wyner [9] showed that 1) $\mathcal{R}^*(D_1, D_2|P_{XY}) \subseteq \mathcal{R}(D_1, D_2|P_{XY})$; 2) $\mathcal{R}(D_1, D_2|P_{XY}) \subseteq \mathcal{R}_{\mathrm{E}}(D_1, D_2|P_{XY})$; 3) $\mathcal{R}_{\mathrm{E}}(D_1, D_2|P_{XY}) \subseteq \mathcal{R}^*(D_1, D_2|P_{XY})$. Thus, these three sets are identical. □

Given private rates $(R_1, R_2) \in \mathbb{R}^2_{+,0}$, define the minimal common rate as

$$\mathsf{R}_0(R_1, R_2, D_1, D_2|P_{XY}) := \inf\big\{r_0 \in \mathbb{R}_{+,0} : (r_0, R_1, R_2) \in \mathcal{R}(D_1, D_2|P_{XY})\big\}. \tag{15}$$

In the next subsection, we introduce the explicit expressions of $\mathsf{R}_0(R_1, R_2, D_1, D_2|P_{XY})$ and $\mathcal{R}(D_1, D_2|P_{XY})$ for the Gaussian Gray-Wyner problem.

B. Existing First-Order Asymptotics

Fix a covariance matrix $\mathbf{\Sigma} = [\Sigma_{jk}]_{(j,k)\in[2]^2} \in \mathbb{S}^2_{++}$ with $\Sigma_{jj} > D_j$ and $\Sigma_{12} > 0$. Let $Q^{\mathbf{\Sigma}}_{XY} := \mathcal{N}(\mathbf{0}_2, \mathbf{\Sigma})$ be the two-dimensional Gaussian distribution and let $(Q^{\mathbf{\Sigma}}_X, Q^{\mathbf{\Sigma}}_Y)$ be the marginal distributions. The Gaussian rate-distortion functions [30, Theorem 10.3.2, Eq. (10.24)] are

$$R_X(Q^{\mathbf{\Sigma}}_X, D_1) = \frac{1}{2}\log\frac{\Sigma_{11}}{D_1}, \tag{16}$$

$$R_Y(Q^{\mathbf{\Sigma}}_Y, D_2) = \frac{1}{2}\log\frac{\Sigma_{22}}{D_2}. \tag{17}$$

Furthermore, define the correlation coefficient:

$$\rho(\mathbf{\Sigma}) := \frac{\Sigma_{12}}{\sqrt{\Sigma_{11}\Sigma_{22}}}, \tag{18}$$

Fix two non-negative real numbers $(r_1, r_2) \in \mathbb{R}^2_{+,0}$. Define

$$\alpha(\mathbf{\Sigma}; r_1, D_1) := \big[R_X(Q^{\mathbf{\Sigma}}_X, D_1) - r_1\big]^+, \tag{19}$$

$$\beta(\mathbf{\Sigma}; r_2, D_2) := \big[R_Y(Q^{\mathbf{\Sigma}}_Y, D_2) - r_2\big]^+. \tag{20}$$

Given a non-negative real number $t \in \mathbb{R}_{+,0}$, define

$$\nu_t := \sqrt{1 - e^{-2t}}. \tag{21}$$

Given any $(\rho, \alpha, \beta) \in (0,1) \times \mathbb{R}^2_{+,0}$, define

$$r_{\alpha,\beta} := (\rho - \nu_\alpha \nu_\beta) e^{\alpha+\beta}. \tag{22}$$

Finally, define the following four sets:

$$\mathcal{P}_1 := \{(\alpha,\beta) \in \mathbb{R}_+^2 : \nu_\alpha \nu_\beta \le \rho \le \min\{\nu_\beta/\nu_\alpha, \nu_\alpha/\nu_\beta\}\}, \tag{23}$$

$$\mathcal{P}_2 := \{(\alpha,\beta) \in \mathbb{R}_+^2 : \rho \le \nu_\alpha \nu_\beta\}, \tag{24}$$

$$\mathcal{P}_3 := \{(\alpha,\beta) \in \mathbb{R}_+^2 : \rho > \nu_\beta/\nu_\alpha\}, \tag{25}$$

$$\mathcal{P}_4 := \{(\alpha,\beta) \in \mathbb{R}_+^2 : \rho > \nu_\alpha/\nu_\beta\}, \tag{26}$$

and the following functions:

$$\Upsilon_\rho(\alpha,\beta) := \begin{cases} \alpha+\beta-\dfrac{1}{2}\log\dfrac{1-r_{\alpha,\beta}^2}{1-\rho^2}, & \text{if } (\alpha,\beta)\in\mathcal{P}_1, \\ \alpha+\beta+\dfrac{1}{2}\log(1-\rho^2), & \text{if } (\alpha,\beta)\in\mathcal{P}_2, \\ \alpha, & \text{if } (\alpha,\beta)\in\mathcal{P}_3, \\ \beta, & \text{if } (\alpha,\beta)\in\mathcal{P}_4 \end{cases} . \tag{27}$$

When $(\alpha,\beta) \in \mathbb{R}_{+,0}^2$ satisfies $\alpha\beta = 0$, define

$$\Upsilon_\rho(\alpha,\beta) := \max\{\alpha,\beta\}. \tag{28}$$

The following lemma is obtained [13, Corollary 2, Theorem 3, Eq. (8)] by rescaling the source parameters to unit variance. Recall that ρ_{XY} is the correlation parameter for the source distribution P_{XY} in (1).

Lemma 2. *For any $(R_1,R_2) \in \mathbb{R}_{+,0}^2$, the minimum common rate satisfies*

$$\mathsf{R}_0(R_1,R_2,D_1,D_2|P_{XY}) = \Upsilon_{\rho_{XY}}\big([R_X(P_X,D_1)-R_1]^+, [R_Y(P_Y,D_2)-R_2]^+\big). \tag{29}$$

It follows from Lemma 2 that the first-order rate region in Definition 2 is

$$\mathcal{R}(D_1,D_2|P_{XY}) = \big\{(R_0,R_1,R_2)\in\mathbb{R}_{+,0}^3 : R_0 \ge \Upsilon_{\rho_{XY}}\big([R_X(P_X,D_1)-R_1]^+, [R_Y(P_Y,D_2)-R_2]^+\big)\big\}. \tag{30}$$

For each fixed private rate pair $(R_1,R_2)\in\mathbb{R}_{+,0}^2$, equality in (30) determines the corresponding minimum common rate.

C. Definitions of Second-Order Asymptotics

To define second-order asymptotics, we need to first specify the boundary rate tuples of the first-order rate-region.

Definition 3. *A first-order rate tuple $\mathbf{r}^* = (R_0^*,R_1^*,R_2^*) \in \mathcal{R}(D_1,D_2|P_{XY}) \subset \mathbb{R}_{+,0}^3$ is a boundary rate tuple if it is Pareto-minimal in the sense that reducing any one of the three rates would make the rate tuple not in the first-order rate region anymore. Specifically, for each $j\in[0:2]$ and any $0<t\le R_j^*$,*

$$\mathbf{r}^* - t\mathbf{e}_{j+1} \notin \mathcal{R}(D_1,D_2|P_{XY}). \tag{31}$$

Recall the source covariance matrix $\mathbf{\Sigma}_{XY}$ in (1), whose diagonal entries are $[\mathbf{\Sigma}_{XY}]_{11} = \sigma_X^2$ and $[\mathbf{\Sigma}_{XY}]_{22} = \sigma_Y^2$. For a first-order rate tuple $\mathbf{r}^* = (R_0^*,R_1^*,R_2^*) \in \mathcal{R}(D_1,D_2|P_{XY}) \subset \mathbb{R}_{+,0}^3$, define

$$(\alpha^*,\beta^*) := \big(R_X(P_X,D_1)-R_1^*, R_Y(P_Y,D_2)-R_2^*\big). \tag{32}$$

The following lemma gives exact conditions for a rate tuple to be a boundary rate tuple.

Lemma 3. *A first-order rate tuple $\mathbf{r}^* = (R_0^*,R_1^*,R_2^*)$ with $R_0^*>0$ is a boundary rate tuple if and only if $R_0^* = \Upsilon_{\rho_{XY}}(\alpha^*,\beta^*)$ and $(\alpha^*,\beta^*)\in\mathbb{R}_+^2$ satisfies one of the following four conditions:*
(a) $(\alpha^*,\beta^*)\in\mathcal{P}_1$, which implies

$$R_0^*+R_1^*+R_2^* = R_X(P_X,D_1)+R_Y(P_Y,D_2)-\frac{1}{2}\log\frac{1-r_{\alpha^*,\beta^*}^2}{1-\rho_{XY}^2}. \tag{33}$$

(b) $(\alpha^*,\beta^*)\in\mathcal{P}_2$, which implies

$$R_0^*+R_1^*+R_2^* = R_X(P_X,D_1)+R_Y(P_Y,D_2)+\frac{1}{2}\log(1-\rho_{XY}^2). \tag{34}$$

(c) $(\alpha^*,\beta^*)\in\mathcal{P}_3$ and $R_2^*=0$, which implies $R_0^*+R_1^* = R_X(P_X,D_1)$.
(d) $(\alpha^*,\beta^*)\in\mathcal{P}_4$ and $R_1^*=0$, which implies $R_0^*+R_2^* = R_Y(P_Y,D_2)$.

The proof of Lemma 3 is provided in Appendix A. Lemma 3 implies that every boundary rate tuple with nonzero common rate belongs to one of the above four regimes. In Fig. 2, we illustrate the boundary rate tuples for a symmetric source with equal distortion levels.

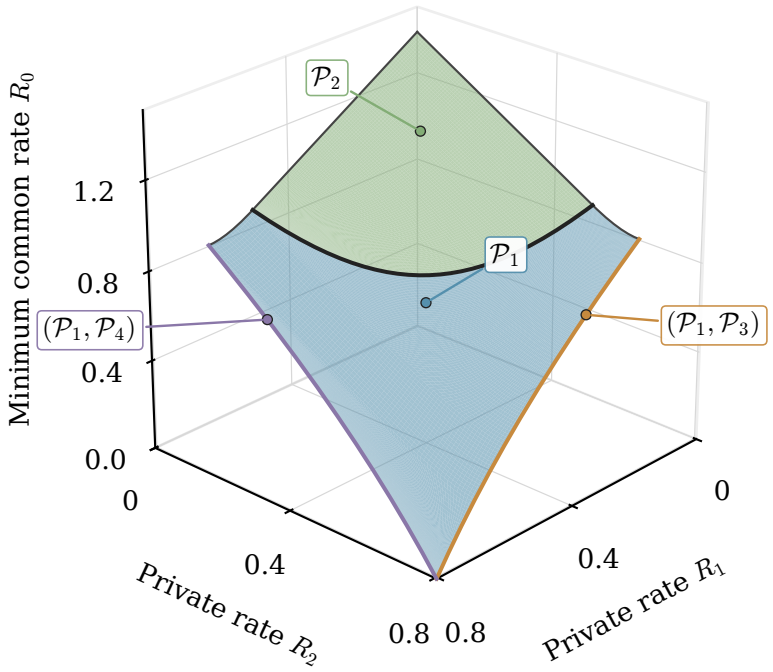


Fig. 2. Boundary rate tuples with positive common rate when $\sigma_X^2 = \sigma_Y^2 = 1$, $D_1 = D_2 = 0.2$, and $\rho_{XY} = 0.6$. The blue and green patches correspond to $(\mathcal{P}_1, \mathcal{P}_2)$ and the boundary of the two regions are the black curve. The orange and purple curves mark the $(\mathcal{P}_1, \mathcal{P}_3)$ and $(\mathcal{P}_1, \mathcal{P}_4)$ interfaces, respectively.

We can now define the second-order asymptotics, which will be characterized in this paper.

Definition 4. *Fix any boundary rate tuple $\mathbf{r}^* = (R_0^*, R_1^*, R_2^*)$ with $R_0^* > 0$ and $\varepsilon \in (0,1)$. A tuple $(L_0, L_1, L_2) \in \mathbb{R}^3$ is said to be second-order $(\mathbf{r}^*, D_1, D_2, \varepsilon)$-achievable if there exists a sequence of (n, M_0, M_1, M_2)-codes such that for each $j \in [0:2]$,*

$$\limsup_{n\to\infty} \frac{\log M_{j,n} - nR_j^*}{\sqrt{n}} \le L_j, \tag{35}$$

and

$$\limsup_{n\to\infty} \varepsilon_n \le \varepsilon. \tag{36}$$

The closure of all second-order $(\mathbf{r}^*, D_1, D_2, \varepsilon)$-achievable tuples is called the $(\mathbf{r}^*, D_1, D_2, \varepsilon)$-achievable second-order region and denoted by $\mathcal{L}(\mathbf{r}^*, D_1, D_2, \varepsilon)$.

Finally, given a boundary rate tuple $\mathbf{r}^* = (R_0^*, R_1^*, R_2^*)$, define the index set for zero private rates as follows:

$$\mathcal{Z}(\mathbf{r}^*) := \{j \in [2] : R_j^* = 0\}. \tag{37}$$

For $j \in \mathcal{Z}(\mathbf{r}^*)$, we would require $L_j \ge 0$ to guarantee that the number of each private message is strictly positive.

III. Second-Order Asymptotics

A. Preliminaries

Fix a boundary rate tuple $\mathbf{r}^* = (R_0^*, R_1^*, R_2^*)$ with $R_0^* > 0$. Recall the definitions of $\mathsf{R}_0(R_1, R_2, D_1, D_2|P_{XY})$ in (15), $\Upsilon_\rho(\alpha, \beta)$ in (27) and (α^*, β^*) in (32). The following lemma proves the differentiability of the minimal common rate.

Proposition 1. *The following results hold.*

(i) The function $(\rho, \alpha, \beta) \mapsto \Upsilon_\rho(\alpha, \beta)$ is continuously differentiable on $(0,1) \times \mathbb{R}_+^2$.

(ii) For every boundary rate tuple $\mathbf{r}^*$ with $R_0^* > 0$, the function $(\mathbf{\Sigma}, r_1, r_2) \mapsto \mathsf{R}_0(r_1, r_2, D_1, D_2|Q_{XY}^{\mathbf{\Sigma}})$ is continuously differentiable in the neighborhood of $(\mathbf{\Sigma}_{XY}, R_1^*, R_2^*)$ in $\mathbb{S}_{++}^2 \times \mathbb{R}^2$. When either private rate equals zero, the derivative is defined from right (above zero).

The proof of Proposition 1 is provided in Appendix B.

Now, let the derivative of $\Upsilon_\rho(\alpha,\beta)$ be

$$(\eta,\lambda_1^*,\lambda_2^*) = \big(\partial_\rho\Upsilon_\rho(\alpha,\beta),\partial_\alpha\Upsilon_\rho(\alpha,\beta),\partial_\beta\Upsilon_\rho(\alpha,\beta)\big)\big|_{(\rho,\alpha,\beta)=(\rho_{XY},\alpha^*,\beta^*)}. \tag{38}$$

It follows from Proposition 1 and Lemma 3 that

$$\mathbf{g} := \nabla_{\mathbf{\Sigma}}\mathsf{R}_0(r_1,r_2,D_1,D_2|Q_{XY}^{\mathbf{\Sigma}})\big|_{(\mathbf{\Sigma},r_1,r_2)=(\mathbf{\Sigma}_{XY},R_1^*,R_2^*)} \tag{39}$$

$$= \frac{1}{2}\begin{bmatrix}(\lambda_1^*-\rho_{XY}\eta)/\sigma_X^2 & \eta/(\sigma_X\sigma_Y)\\ \eta/(\sigma_X\sigma_Y) & (\lambda_2^*-\rho_{XY}\eta)/\sigma_Y^2\end{bmatrix}. \tag{40}$$

$$g_i := \partial_{r_i}\mathsf{R}_0(r_1,r_2,D_1,D_2|Q_{XY}^{\mathbf{\Sigma}})\big|_{(\mathbf{\Sigma},r_1,r_2)=(\mathbf{\Sigma}_{XY},R_1^*,R_2^*)} = -\lambda_i^*, \qquad i\in[2]. \tag{41}$$

Let $\boldsymbol{\lambda}^* := (\lambda_1^*,\lambda_2^*)\in[0,1]^2$. For any $(x,y)\in\mathbb{R}^2$, let $\mathbf{x} := (x,y)^\mathsf{T}$ and define the following information density:

$$\jmath(x,y|R_1^*,R_2^*,D_1,D_2,P_{XY}) := R_0^* + \mathbf{x}^\mathsf{T}\mathbf{g}\mathbf{x} - \mathrm{tr}(\mathbf{g}\mathbf{\Sigma}_{XY}). \tag{42}$$

We remark that $\jmath(x,y|R_1^*,R_2^*,D_1,D_2,P_{XY})$ is the distortion-tilted information density for the Gaussian Gray-Wyner problem, analogous to the discrete case in [11, Definition 4, Eqs. (24)–(26)]. Furthermore, define the dispersion, the variance of the distortion-tilted information density, as follows:

$$V(R_1^*,R_2^*,D_1,D_2|P_{XY}) := \mathsf{Var}_{P_{XY}}[\jmath(X,Y|R_1^*,R_2^*,D_1,D_2,P_{XY})]. \tag{43}$$

For ease of notation, we use $(\jmath(x,y),V)$ to denote $(\jmath(x,y|R_1^*,R_2^*,D_1,D_2,P_{XY}),V(R_1^*,R_2^*,D_1,D_2|P_{XY}))$. Recall the definitions of $\nu_\cdot$ in (21). Define

$$u := \frac{\nu_{\beta^*}}{\nu_{\alpha^*}}\exp\big(\beta^*-\alpha^*\big) = \sqrt{\frac{1-e^{-2\beta^*}}{1-e^{-2\alpha^*}}}\exp\big(\beta^*-\alpha^*\big). \tag{44}$$

The following lemma presents properties of the distortion-tilted information density and the dispersion function.

Lemma 4. *For the fixed boundary tuple $\mathbf{r}^*$ with $R_0^*>0$, the following claims hold.*

(i) The distortion-tilted information density satisfies

$$\mathsf{E}_{P_{XY}}[\jmath(X,Y)] = R_0^*. \tag{45}$$

(ii) The dispersion function satisfies

$$V = 2\,\mathrm{tr}[(\mathbf{g}\mathbf{\Sigma}_{XY})^2] = \frac{1+(\lambda_1^*+\lambda_2^*-1)^2}{2}. \tag{46}$$

In particular, the dispersion function has the following explicit form:

$$V = \begin{cases} 0.5\left[1+\left(\dfrac{(u-r)(1-ru)}{u(1-r^2)}\right)^2\right], & \text{if } (\alpha^*,\beta^*)\in\mathcal{P}_1,\\ 1, & \text{if } (\alpha^*,\beta^*)\in\mathcal{P}_2,\\ 0.5, & \text{if } (\alpha^*,\beta^*)\in\mathcal{P}_3,\\ 0.5, & \text{if } (\alpha^*,\beta^*)\in\mathcal{P}_4\end{cases}. \tag{47}$$

The proof of Lemma 4 is provided in Appendix C.

B. Main Results and Discussions

Recall that the source distribution is $P_{XY}=\mathcal{N}(\mathbf{0}_2,\mathbf{\Sigma}_{XY})$. Furthermore, recall the definitions of the second-order rate region $\mathcal{L}(\mathbf{r}^*,D_1,D_2,\varepsilon)$ in Definition 4 and the zero private rate index set $\mathcal{Z}(\mathbf{r}^*)$ in (37). Fix any boundary rate tuple $\mathbf{r}^*=(R_0^*,R_1^*,R_2^*)$ with $R_0^*>0$.

Theorem 1. *Given any $\varepsilon\in(0,1)$, the second-order rate region satisfies*

$$\mathcal{L}(\mathbf{r}^*,D_1,D_2,\varepsilon) = \Big\{(L_0,L_1,L_2)\in\mathbb{R}^3 : L_0+\lambda_1^*L_1+\lambda_2^*L_2 \ge \sqrt{V}\,\mathrm{Q}^{-1}(\varepsilon),\quad L_j\ge 0\ \forall j\in\mathcal{Z}(\mathbf{r}^*)\Big\}. \tag{48}$$

The achievability and converse proofs of Theorem 1 are provided in Sections IV and V, respectively. We provide a proof sketch here. In the achievability part, we group correlated source sequences into power shell types according to their empirical covariance and derive a covering lemma that bounds the common rate required to cover each power shell type with desired private rates. Specifically, the covariance shell covering lemma enables us to apply the Berry-Esseen theorem to obtain the desired second-order asymptotics. In the converse part, we use the existing result from [12, Theorem 17] when $(\lambda_1^*,\lambda_2^*)$ are both

positive and use the converse result of the P2P case [6, Theorem 40] when either λ_1^* or λ_2^* equals zero. However, the application of [12, Theorem 17] requires one to relate the entropy support expression in [12, Theorem 17] with the distortion-tilted information density and the dispersion function.

Theorem 1 shows that the second-order rate region for the Gaussian Gray-Wyner problem has exactly the same form as the discrete counterpart [11, Theorem 4]. In the following corollary, we explicitly calculate the second-order rate region for the Gaussian Gray-Wyner problem. Recall the definitions of $r_{\cdot}$ in (22), (α^*, β^*) in (32), the four sets $(\mathcal{P}_1, \ldots, \mathcal{P}_4)$ in (23)–(26), and u in (44). For ease of notation, we use r to denote r_{α^*,β^*}.

Corollary 2. *The second-order rate region consists of rate tuples (L_0, L_1, L_2) such that $L_j \geq 0$ for $j \in \mathcal{Z}(\mathbf{r}^*)$ and has the following explicit form depending on parameters (α^*, β^*) that are defined via the private rates (R_1^*, R_2^*) of the boundary first-order rate tuple $r^* = (R_0^*, R_1^*, R_2^*)$.*

(i) If $(\alpha^*, \beta^*) \in \mathcal{P}_1$,

$$L_0 + \frac{1 - ru}{1 - r^2} L_1 + \frac{1 - r/u}{1 - r^2} L_2 \geq \sqrt{V}\, \mathrm{Q}^{-1}(\varepsilon). \tag{49}$$

(ii) If $(\alpha^*, \beta^*) \in \mathcal{P}_2$,

$$L_0 + L_1 + L_2 \geq \mathrm{Q}^{-1}(\varepsilon). \tag{50}$$

(iii) If $(\alpha^*, \beta^*) \in \mathcal{P}_3$,

$$L_0 + L_1 \geq \frac{\mathrm{Q}^{-1}(\varepsilon)}{\sqrt{2}}. \tag{51}$$

(iv) If $(\alpha^*, \beta^*) \in \mathcal{P}_4$,

$$L_0 + L_2 \geq \frac{\mathrm{Q}^{-1}(\varepsilon)}{\sqrt{2}}. \tag{52}$$

The neighboring conditions agree at every finite interface. In particular, the one-source expressions also hold at their adjoining $\mathcal{P}_1$ interfaces.

The proof of Corollary 2 is provided in Appendix H.

IV. Achievability Proof

The achievability proof here is parallel to the discrete Gray–Wyner argument [11, Section IV-A]. In particular, we replace the type covering lemma tailored to the discrete source by a covariance shell covering lemma that is judiciously derived for the bivariate Gaussian source. This lemma is of independent interest for other multiterminal lossy source coding problems concerning correlated Gaussian sources.

A. Preliminaries

Fix any integer $n \in \mathbb{N}$ and $\mathbf{x}^n = (x^n, y^n) \in (\mathbb{R}^n)^2$. Define the following empirical second-moment matrix:

$$\hat{\mathbf{\Sigma}}(\mathbf{x}^n) := \frac{1}{n} \mathbf{x}^n (\mathbf{x}^n)^{\mathsf{T}}. \tag{53}$$

We remark that the diagonal elements of $\hat{\mathbf{\Sigma}}(\mathbf{x}^n)$ are empirical powers $\|x^n\|^2/n$ and $\|y^n\|^2/n$. In what follows, we will use $\hat{\mathbf{\Sigma}}$ to denote the random covariance matrix $\hat{\mathbf{\Sigma}}(\mathbf{X}^n)$ when $\mathbf{X}^n = (X^n, Y^n)$ denotes the random source sequences. Recall from (1) that the covariance matrix is $\mathbf{\Sigma}_{XY}$, the standard deviations are (σ_X, σ_Y) and ρ_{XY} is the correlation parameter of the source distribution. Let $\lambda_{\min}(\mathbf{\Sigma}_{XY})$ be its smallest eigenvalue. Define the following two constants

$$c_{\mathrm{cor}} := \sigma_X^{-2} + \sigma_Y^{-2} + \frac{2}{\sigma_X \sigma_Y}, \tag{54}$$

$$\eta_{\mathrm{cov}} := \frac{1}{8} \min \left\{ \lambda_{\min}(\mathbf{\Sigma}_{XY}), \sigma_X^2 - D_1, \sigma_Y^2 - D_2, \frac{\rho_{XY}}{c_{\mathrm{cor}}}, \frac{1 - \rho_{XY}}{c_{\mathrm{cor}}} \right\}. \tag{55}$$

Furthermore, define the two sets:

$$\mathcal{K}_0 := \{\mathbf{\Sigma} \in \mathbb{S}^2 : \|\mathbf{\Sigma} - \mathbf{\Sigma}_{XY}\| \leq \eta_{\mathrm{cov}}\}, \tag{56}$$

$$\mathcal{K} := \{\mathbf{\Sigma} \in \mathbb{S}^2 : \|\mathbf{\Sigma} - \mathbf{\Sigma}_{XY}\| \leq 2\eta_{\mathrm{cov}}\}. \tag{57}$$

Given a real number $t \in \mathbb{R}$, define the quantization function $\mathfrak{r}_n(t) := \lfloor nt + 1/2 \rfloor / n$. Given any matrix $\mathbf{\Sigma} \in \mathbb{S}^2$, define its quantized version by

$$\mathfrak{q}_n(\mathbf{\Sigma}) := \begin{bmatrix} \mathfrak{r}_n(\Sigma_{11}) & \mathfrak{r}_n(\Sigma_{12}) \\ \mathfrak{r}_n(\Sigma_{12}) & \mathfrak{r}_n(\Sigma_{22}). \end{bmatrix} \tag{58}$$

Let

$$\mathfrak{q}_n(\mathcal{K}_0) := \{\mathfrak{q}_n(\mathbf{\Sigma}) : \ \Sigma \in \mathcal{K}_0\} \tag{59}$$

be the set of distinct quantized versions for the matrices in the set $\mathcal{K}_0$ and let $N_n := |\mathfrak{q}_n(\mathcal{K}_0)|$ denote its size. We also write $\mathfrak{q}_n(\mathcal{K}_0)$ as $(\Sigma_1, \ldots, \Sigma_{N_n})$. For each $\Sigma \in \mathcal{K}_0$,

$$\|\mathfrak{q}_n(\mathbf{\Sigma}) - \mathbf{\Sigma}\|^2 = |\mathfrak{r}_n(\Sigma_{11}) - \Sigma_{11}|^2 + 2|\mathfrak{r}_n(\Sigma_{12}) - \Sigma_{12}|^2 + |\mathfrak{r}_n(\Sigma_{22}) - \Sigma_{22}|^2, \tag{60}$$

$$\leq \frac{1+2+1}{4n^2} = \frac{1}{n^2}, \tag{61}$$

where each scalar rounding error is at most $\frac{1}{2n}$. Thus, $\|\mathfrak{q}_n(\mathbf{\Sigma}) - \mathbf{\Sigma}\| \leq 1/n$. The triangle inequality implies $\mathfrak{q}_n(\mathcal{K}_0) \subset \mathcal{K}$ when $\frac{1}{n} \leq \eta_{\mathrm{cov}}$. Since each entry of $\mathfrak{q}_n(\mathbf{\Sigma})$ and $\mathfrak{q}_n(\mathbf{\Sigma})$ differs by at most $\frac{1}{n}$ for any $\Sigma \in \mathcal{K}_0$, there are at most $\lfloor 2n\eta_{\mathrm{cov}} \rfloor + 3$ different quantization values in the range of $2\eta_{\mathrm{cov}}$. As a result,

$$N_n \leq (\lfloor 2n\eta_{\mathrm{cov}} \rfloor + 3)^3 \leq (2\eta_{\mathrm{cov}} + 3)^3 n^3. \tag{62}$$

Recall that $\mathbf{x}^n = (x^n, y^n) \in (\mathbb{R}^n)^2$. Given a constant $C_0 \geq 1$ and a matrix $\mathbf{\Sigma} \in \mathcal{K}$, define the following set of source sequences:

$$\mathcal{T}_n(\mathbf{\Sigma}) := \left\{ \mathbf{x}^n \in \mathbb{R}^{2\times n} : \left\| \hat{\mathbf{\Sigma}}(\mathbf{x}^n) - \mathbf{\Sigma} \right\| \leq \frac{C_0}{n} \right\}. \tag{63}$$

For each $q \in [N_n]$, define

$$\mathcal{T}_{n,q} := \left\{ \mathbf{x}^n \in \mathbb{R}^{2\times n} : \hat{\mathbf{\Sigma}}(\mathbf{x}^n) \in \mathcal{K}_0, \ \mathfrak{q}_n(\hat{\mathbf{\Sigma}}(\mathbf{x}^n)) = \mathbf{\Sigma}_q \right\}. \tag{64}$$

It follows from (61) that $\mathcal{T}_{n,q} \subseteq \mathcal{T}_n(\mathbf{\Sigma}_q)$. We remark that $\mathcal{T}_n(\mathbf{\Sigma})$ is analogous to the typical set while $\mathcal{T}_{n,q}$ is analogous to the type class. Recall the definition of the minimum common rate in (29).

Fix non-negative real numbers $(r_0, r_1, r_2) \in \mathbb{R}^3_{+,0}$, define the index set

$$\mathcal{J}_n := \{q \in [N_n] : \ \mathsf{R}_0(r_1, r_2, D_1, D_2 | Q_{XY}^{\mathbf{\Sigma}_q}) \leq r_0\}. \tag{65}$$

For each $q \in [N_n]$, define the following set of covariance matrices:

$$\mathcal{F}_q := \left\{ \mathbf{\Sigma}_{\mathrm{N}} \in \mathbb{S}^2_+ : \mathbf{\Sigma}_{\mathrm{N}} \preceq \mathbf{\Sigma}_q, \ \Sigma_{\mathrm{N},jj} \leq D_j e^{2r_j} \ \forall \ j \in [2] \right\}. \tag{66}$$

Furthermore, let

$$\mathbf{\Sigma}_{\mathrm{N},q} := \operatorname*{arg\,max}_{\mathbf{\Sigma}_{\mathrm{N}} \in \mathcal{F}_q} |\mathbf{\Sigma}_{\mathrm{N}}| \tag{67}$$

be the covariance matrix in $\mathcal{F}_q$ that has maximum determinant. Finally, when $n \geq 3$, define

$$s_n := \frac{\sqrt{n}}{\log n}, \tag{68}$$

and give real number $\kappa \in (0, 1)$, define a parameter

$$\chi_\kappa := 2 + 2\sqrt{\kappa^{-1} + 1}. \tag{69}$$

The definitions above will be used to present our coding scheme in the next subsection.

B. Coding Scheme

1) Codebook Generation: For each $q \in \mathcal{J}_n$, we generate one common codebook and two private codebooks. The encoder and both decoders know these codebooks and the quantized centers $\mathfrak{q}_n(\mathcal{K}_0) = (\Sigma_1, \ldots, \Sigma_{N_n})$ before transmission.

Fix $q \in \mathcal{J}_n$. Define the following constant

$$M_{0,q} := \left\lceil \exp\left\{ \frac{n}{2} \log \frac{|\mathbf{\Sigma}_q|}{|\mathbf{\Sigma}_{\mathrm{N},q}|} + \frac{s_n}{8} \right\} \right\rceil. \tag{70}$$

For each $m_0 \in [M_{0,q}]$, generate $\mathbf{u}_q^n(m_0) \in \mathbb{R}^{2\times n}$ i.i.d. from the two-dimensional Gaussian distribution $\mathcal{N}(\mathbf{0}_2, \mathbf{1}_{2\times 2})$ columns and let

$$\mathbf{w}_q^n(m_0) := (\mathbf{\Sigma}_q - \mathbf{\Sigma}_{\mathrm{N},q})^{1/2} \mathbf{u}_q^n(m_0). \tag{71}$$

Note that each column of $\mathbf{w}_q^n(m_0)$ has distribution $\mathcal{N}(\mathbf{0}_2, \mathbf{\Sigma}_q - \mathbf{\Sigma}_{\mathrm{N},q})$. For each $j \in [2]$, define the following constant:

$$\xi_{j,q,n} := \sqrt{n[\mathbf{\Sigma}_{\mathrm{N},q}]_{jj} + \chi_\kappa s_n}. \tag{72}$$

The next lemma shows that there exists deterministic private codebooks that cover two source sequences within desired distortion levels, which is proved in Appendix D.

Lemma 5. *Given any $\kappa \in (0,1)$, there exists a constant $C_{\mathrm{cov}} \geq 1$, depending only on (κ, D_1, D_2), such that the following result holds. For each $n \geq 3$, and $q \in \mathcal{J}_n$ satisfying $\mathbf{\Sigma}_{\mathrm{N},q} \preceq \kappa^{-1}\mathbf{1}_2$, and each $j \in [2]$, there exists a deterministic private codebook $\{v_{j,q}^n(m_j) : m_j \in [M_{j,q}]\} \subset \mathbb{R}^n$ such that*

$$M_{j,q} \leq C_{\mathrm{cov}} n^4 \max\left\{ 1, \left(\frac{\xi_{j,q,n}^2}{nD_j} \right)^{n/2} \right\}, \tag{73}$$

$$\mathcal{B}_n(\mathbf{0}_n, \xi_{j,q,n}) \subseteq \bigcup_{m_j \in [M_{j,q}]} \mathcal{B}_n(v_{j,q}^n(m_j), \sqrt{nD_j}). \tag{74}$$

2) Encoding and Decoding: We now specify the encoders and decoders using codebooks $\{\mathbf{w}_q^n(m_0) : \ m_0 \in [M_{0,q}], \ q \in \mathcal{J}_n\}$, $\{v_{1,q}^n(m_1) : \ m_1 \in [M_{1,q}]\}$ and $\{v_{2,q}^n(m_2) : \ m_2 \in [M_{2,q}]\}$. Recall that $\mathbf{x}^n = (x^n, y^n)$, $(\mathbf{x}^n)_1 = x^n$ and $(\mathbf{x}^n)_2 = y^n$. For each $q \in \mathcal{J}_n$, $m_0 \in [M_{0,q}]$, and $j \in [2]$, define the following sets

$$\mathcal{A}_{0,n,q}(\mathbf{x}^n) := \{m_0 \in [M_{0,q}] : \left\| (\mathbf{x}^n)_j - \mathbf{w}_q^n(m_0)_j \right\|^2 \leq \xi_{j,q,n}^2, \ \forall \ j \in [2]\}, \tag{75}$$

$$\mathcal{A}_{j,n,q}(\mathbf{x}^n, m_0) := \{m_j \in [M_{j,q}] : \left\| (\mathbf{x}^n)_j - \mathbf{w}_q^n(m_0)_j - v_{j,q}^n(m_j) \right\|^2 \leq nD_j\}. \tag{76}$$

The set $\mathcal{A}_{0,n,q}(\mathbf{x}^n)$ collects the indices of common codewords such that the distortion between each source sequence and the common codeword is within distortion level that can be covered by private codewords in Lemma 5 while the set $\mathcal{A}_{j,n,q}(\mathbf{x}^n, m_0)$ collects indices of private codewords that cover the j-th source sequence within target distortion levels.

Let $\mathcal{H}_{n,q} := \{\mathbf{x}^n : \mathcal{A}_{0,n,q}(\mathbf{x}^n) \neq \emptyset\}$. Let 0 be the fallback common symbol and define the following alphabet:

$$\mathcal{M}_{0,n} := \{0\} \cup \Big(\cup_{q \in \mathcal{J}_n} (\{q\} \times [M_{0,q}]) \Big), \tag{77}$$

whose size is

$$M_0 := 1 + \sum_{q \in \mathcal{J}_n} M_{0,q}. \tag{78}$$

Furthermore, for each $j \in [2]$, define

$$M_j := \begin{cases} \max\limits_{q \in \mathcal{J}_n} M_{j,q}, & \text{if } \mathcal{J}_n \neq \varnothing, \\ 1, & \text{if } \mathcal{J}_n = \varnothing. \end{cases} \tag{79}$$

Let $\iota_n : \mathcal{M}_{0,n} \to [M_0]$ be a fixed bijection with $\iota_n(0) = 1$, known to the encoder and both decoders. This function enables us to assign a single common message for each pair $(q, m_0) \in \mathcal{J}_n \times [M_{0,q}]$. For each $q \in \mathcal{J}_n$, define an encoder $f_q = (f_{0,q}, f_{1,q}, f_{2,q})$ such that if $\mathbf{x}^n \in \mathcal{H}_{n,q}$, $f_{0,q}(\mathbf{x}^n)$ is the smallest index in $\mathcal{A}_{0,n,q}(\mathbf{x}^n)$ and, for each $j \in [2]$, $f_{j,q}(\mathbf{x}^n)$ is the smallest index in $\mathcal{A}_{j,n,q}(\mathbf{x}^n, f_{0,q}(\mathbf{x}^n))$; otherwise, $f_q(\mathbf{x}^n) = (1,1,1)$. Now, define the encoder $f = (f_0, f_1, f_2)$ as follows:

$$f(\mathbf{x}^n) := \begin{cases} \big(\iota_n(q, f_{0,q}(\mathbf{x}^n)), f_{1,q}(\mathbf{x}^n), f_{2,q}(\mathbf{x}^n)\big), & \text{if } \mathbf{x}^n \in \mathcal{T}_{n,q} \cap \mathcal{H}_{n,q}, \text{ for some } q \in \mathcal{J}_n, \\ \big(\iota_n(0), 1, 1\big), & \text{otherwise} \end{cases}. \tag{80}$$

We next specify the decoder operation. Fix any $j \in [2]$. For each $q \in \mathcal{J}_n$, define

$$\phi_{j,q}(m_0, m_j) := \mathbf{w}_q^n(m_0)_j + v_{j,q}^n(m_j). \tag{81}$$

Given a common message other than $\iota_n(0)$, decoder ϕ_j uses the inverse function ι_n^{-1} to obtain (q, m_0) use compresses messages (m_0, m_j) to generate the source sequence estimate as follows:

$$\phi_j(\iota_n(q, m_0), m_j) := \phi_{j,q}(m_0, m_j). \tag{82}$$

If the common message equals $\iota_n(0)$, each decoder outputs source sequence estimate as $\mathbf{0}_n$.

C. Covariance Covering and Finite Blocklength Bound

Recall from (59) that $\mathfrak{q}_n(\mathcal{K}_0) = (\Sigma_1, \ldots, \Sigma_{N_n})$ and the definitions of $\mathcal{F}_q$ in (66) and the optimizer $\mathbf{\Sigma}_{\mathrm{N},q}$ in (67). The next lemma applies directly to the cell codebooks and local maps defined above. The following lemma bounds the message sizes required to ensure that the reproduction of our coding scheme satisfies the distortion constraints for both decoders, which named covariance shell covering lemma.

Lemma 6. *There exist constants $\kappa \in (0,1)$ and $C \in \mathbb{R}_+$, and an integer $n_0 \in \mathbb{N}$ such that for every $n \geq n_0$ and each $q \in \mathcal{J}_n$, $\mathbf{x}^n \in \mathcal{T}_n(\mathbf{\Sigma}_q)$ and $j \in [2]$,*

$$d\big((\mathbf{x}^n)_j, \phi_{j,q}(f_{0,q}(\mathbf{x}^n), f_{j,q}(\mathbf{x}^n))\big) \leq D_j, \tag{83}$$

$$\log M_{0,q} \leq \frac{n}{2} \log \frac{|\mathbf{\Sigma}_q|}{|\mathbf{\Sigma}_{\mathrm{N},q}|} + C s_n, \tag{84}$$

$$\log M_{j,q} \leq \frac{n}{2} \left[\log \frac{[\mathbf{\Sigma}_{\mathrm{N},q}]_{jj}}{D_j}\right]^+ + C s_n + C \log n. \tag{85}$$

The proof of Lemma 6 is provided in Appendix E.

Fix three non-negative real numbers $(r_0, r_1, r_2) \in \mathbb{R}^3_{+,0}$. Recall the definitions of the joint excess-distortion probability ε_n in (8), and s_n in (68). Let C be the constant in Lemma 6 and define $h_n := C(s_n + \log n)$. The following lemma upper bounds the excess-distortion probability of our coding scheme.

Lemma 7. *For sufficiently large n, there exists an (n, M_0, M_1, M_2)-code per Def. 1 such that for each $j \in [0:2]$,*

$$\log M_j \leq n r_j + h_n, \tag{86}$$

and

$$\varepsilon_n \leq P_{XY}^n\{\hat{\mathbf{\Sigma}}_n \notin \mathcal{K}_0\} + P_{XY}^n\{\hat{\mathbf{\Sigma}}_n \in \mathcal{K}_0,\ \mathsf{R}_0(r_1, r_2, D_1, D_2 | Q_{XY}^{\hat{\mathbf{\Sigma}}_n}) > r_0\}. \tag{87}$$

Proof. We first bound the excess-distortion probability. It follows from our coding scheme and Lemma 6 that the joint excess-distortion probability satisfies

$$\varepsilon_n \leq P_{XY}^n\{\hat{\mathbf{\Sigma}}_n \notin \mathcal{K}_0\} + \sum_{q \in [N_n] \setminus \mathcal{J}_n} P_{XY}^n\{\hat{\mathbf{\Sigma}}_n \in \mathcal{K}_0, \mathfrak{q}_n(\hat{\mathbf{\Sigma}}_n) = \mathbf{\Sigma}_q\}, \tag{88}$$

$$= P_{XY}^n\{\hat{\mathbf{\Sigma}}_n \notin \mathcal{K}_0\} + P_{XY}^n\{\hat{\mathbf{\Sigma}}_n \in \mathcal{K}_0, \mathsf{R}_0(r_1, r_2, D_1, D_2 | Q_{XY}^{\hat{\mathbf{\Sigma}}_n}) > r_0\}, \tag{89}$$

We next show that the rate constraints are satisfied. For each $q \in \mathcal{J}_n$, it follows from the Gaussian first-order characterization in [13, Theorem 3 and Corollary 2] and the covariance representation verified in Appendix E that

$$\begin{aligned} \mathsf{R}_0(r_1, r_2, D_1, D_2 | Q_{XY}^{\mathbf{\Sigma}_q}) &= \min_{\substack{\mathbf{\Sigma}_{\mathrm{N}} \in \mathcal{F}_q \\ \mathbf{\Sigma}_{\mathrm{N}} \succ 0}} \frac{1}{2} \log \frac{|\mathbf{\Sigma}_q|}{|\mathbf{\Sigma}_{\mathrm{N}}|} \\ &= \frac{1}{2} \log \frac{|\mathbf{\Sigma}_q|}{|\mathbf{\Sigma}_{\mathrm{N},q}|} \leq r_0, \end{aligned} \tag{90}$$

where the second equality follows from (67), and the inequality follows since $q \in \mathcal{J}_n$ (cf. (65)). Furthermore, it follows from the definition of the optimizer $\mathbf{\Sigma}_{\mathrm{N},q}$ in (67), in particular, the constraint $\mathbf{\Sigma}_{\mathrm{N},q} \in \mathcal{F}_q$ (cf. (66)) that for each $j \in [2]$,

$$\frac{1}{2} \left[\log \frac{[\mathbf{\Sigma}_{\mathrm{N},q}]_{jj}}{D_j}\right]^+ \leq r_j. \tag{91}$$

Recall from our coding scheme that if $\mathcal{J}_n = \varnothing$, we have $M_0 = M_1 = M_2 = 1$, and rate constraints are satisfied. In the following, we consider the case $\mathcal{J}_n \neq \varnothing$. It follows from Lemma 6 that

$$M_0 \leq 1 + N_n e^{n r_0 + C s_n} \leq \exp\big(n r_0 + C s_n + C \log n\big), \tag{92}$$

where the first inequality follows from the results in (84) and (90), and the second inequality follows from (62) which implies that $N_n \leq C n^3$. Furthermore, for each $j \in [2]$, it follows from (91) and (85) that

$$M_j \leq \exp\big(n r_j + C s_n + C \log n\big). \tag{93}$$

The rate constraints are now verified. □

D. Wrap Up and Final Steps

Fix a boundary rate tuple $\mathbf{r}^* = (R_0^*, R_1^*, R_2^*)$ with $R_0^* > 0$. Define $b_n := (\log n)^{-1/4}$ and

$$r_{0,n} := R_0^* + \frac{L_0 + b_n}{\sqrt{n}}, \tag{94}$$

$$r_{j,n} := R_j^* + \frac{L_j}{\sqrt{n}},\ j \in [2]. \tag{95}$$

For all sufficiently large n, the maximum is inactive when $R_j^* > 0$. When $R_j^* = 0$, the condition $L_j \geq 0$ in Theorem 1 implies $r_{j,n} = L_j/\sqrt{n}$, which equals zero if $L_j = 0$.

The above maximum is inactive at positive private rates and equals $L_j/\sqrt{n}$ when $R_j^* = 0$. For all sufficiently large n and each $j \in [0:2]$, it follows from Lemma 7 with $(r_0, r_1, r_2) = (r_{0,n}, r_{1,n}, r_{2,n})$ that

$$\frac{\log M_{j,n} - nR_j^*}{\sqrt{n}} \leq L_j + b_n + \frac{h_n}{\sqrt{n}}. \tag{96}$$

Since $b_n \to 0$ and $h_n = o(\sqrt{n})$, it follows that for each $j \in [0:2]$,

$$\limsup_{n\to\infty} \frac{\log M_{j,n} - nR_j^*}{\sqrt{n}} \leq L_j. \tag{97}$$

Furthermore, substituting (94) and (95) into (87) leads to

$$\varepsilon_n \leq P_{XY}^n\{\hat{\mathbf{\Sigma}}_n \notin \mathcal{K}_0\} + P_{XY}^n\left\{\hat{\mathbf{\Sigma}}_n \in \mathcal{K}_0, \sqrt{n}[\mathsf{R}_0(r_{1,n}, r_{2,n}, D_1, D_2|Q_{XY}^{\hat{\mathbf{\Sigma}}_n}) - R_0^*] > L_0 + b_n\right\}. \tag{98}$$

In the remaining proof, it suffices to show that $\varepsilon_n \leq \varepsilon$ asymptotically.

We now bound the first term in (98). Recall that $\hat{\mathbf{\Sigma}}_n = \hat{\mathbf{\Sigma}}_n(\mathbf{X}^n)$ and $\mathbf{X}^n = (X^n, Y^n)$ denotes the random source sequences. It follows from the memoryless nature of the source that

$$\mathsf{E}_{P_{XY}^n}\left[\left\|\hat{\mathbf{\Sigma}}_n - \mathbf{\Sigma}_{XY}\right\|^2\right] = \frac{(\operatorname{tr}\mathbf{\Sigma}_{XY})^2 + \operatorname{tr}(\mathbf{\Sigma}_{XY}^2)}{n}. \tag{99}$$

Furthermore, it follows from the the definition of $\mathcal{K}_0$ in (56) and Markov's inequality that

$$P_{XY}^n\{\hat{\mathbf{\Sigma}}_n \notin \mathcal{K}_0\} \leq \frac{(\operatorname{tr}\mathbf{\Sigma}_{XY})^2 + \operatorname{tr}(\mathbf{\Sigma}_{XY}^2)}{n\eta_{\mathrm{cov}}^2} \tag{100}$$

$$= \frac{2\sigma_X^4 + 2\sigma_Y^4 + 2\sigma_X^2\sigma_Y^2(1+\rho_{XY}^2)}{n\eta_{\mathrm{cov}}^2} = O\left(\frac{1}{n}\right). \tag{101}$$

We next bound the second term of (98). Recall that $\mathbf{X} = (X, Y)^{\mathsf{T}}$. It follows from the definition of the distortion-tilted information density $\jmath(\cdot)$ in (42) that

$$\mathsf{E}_{Q_{XY}^{\hat{\mathbf{\Sigma}}_n}}[\jmath(X,Y)] = R_0^* + \mathsf{E}_{Q_{XY}^{\hat{\mathbf{\Sigma}}_n}}[\operatorname{tr}(\mathbf{g}\mathbf{X}\mathbf{X}^{\mathsf{T}})] - \operatorname{tr}(\mathbf{g}\mathbf{\Sigma}_{XY}) \tag{102}$$

$$= R_0^* + \operatorname{tr}(\mathbf{g}\hat{\mathbf{\Sigma}}_n) - \operatorname{tr}(\mathbf{g}\mathbf{\Sigma}_{XY}) \tag{103}$$

$$= R_0^* + \left\langle \mathbf{g}, \hat{\mathbf{\Sigma}}_n - \mathbf{\Sigma}_{XY}\right\rangle, \tag{104}$$

where the first equality follows since $\mathbf{X}^{\mathsf{T}}\mathbf{g}\mathbf{X} = \operatorname{tr}(\mathbf{g}\mathbf{X}\mathbf{X}^{\mathsf{T}})$, the second equality interchanges a finite sum with expectation, and the last equality uses the symmetry of $\mathbf{g}$. It follows from Proposition 1 that we have have the following Taylor's expansion positive definite matrix $\mathbf{\Sigma}$:

$$\mathsf{R}_0(r_1, r_2, D_1, D_2|Q_{XY}^{\hat{\mathbf{\Sigma}}_n}) = R_0^* + \left\langle \mathbf{g}, \hat{\mathbf{\Sigma}}_n - \mathbf{\Sigma}_{XY}\right\rangle - \sum_{j\in[2]} \lambda_j^*(r_j - R_j^*) + o\Big(\left\|\hat{\mathbf{\Sigma}}_n - \mathbf{\Sigma}_{XY}\right\| + \sum_{j\in[2]} |r_j - R_j^*|\Big). \tag{105}$$

Define

$$\mathsf{A}_n := \sqrt{n}\left\langle \mathbf{g}, \hat{\mathbf{\Sigma}}_n - \mathbf{\Sigma}_{XY}\right\rangle, \tag{106}$$

$$= \frac{1}{\sqrt{n}} \sum_{i\in[n]} \left[\mathbf{X}_i^{\mathsf{T}}\mathbf{g}\mathbf{X}_i - \operatorname{tr}(\mathbf{g}\mathbf{\Sigma}_{XY})\right], \tag{107}$$

$$= \frac{1}{\sqrt{n}} \sum_{i\in[n]} \left[\jmath(X_i, Y_i) - R_0^*\right], \tag{108}$$

where Eq. (107) follows from (53), and Eq. (108) follows from (42). Replacing $\hat{\mathbf{\Sigma}}_n$ by its quantization would destroy this exact representation as a sum of independent random variables.

Define the third absolute moment by $\beta_3 := \mathsf{E}_{P_{XY}}[|\jmath(X,Y) - R_0^*|^3]$. Since $\jmath(X,Y)$ in (42) is a quadratic function of a Gaussian vector,

$$\beta_3 \le 4\,\|\mathbf{g}\|^3\,\mathsf{E}_{P_{XY}}[\|\mathbf{X}\|^6] + 4|\operatorname{tr}(\mathbf{g}\mathbf{\Sigma}_{XY})|^3 < \infty, \tag{109}$$

where the last inequality follows since Gaussian random variables have finite sixth moments. Invoking the Berry–Esseen theorem [16, Theorem 1.1], we conclude that there exists a finite positive universal constant C_{BE} such that

$$\sup_{t\in\mathbb{R}}\left|P_{XY}^n\{\mathsf{A}_n \le t\} - \Phi(t/\sqrt{V})\right| \le \frac{C_{\mathrm{BE}}\beta_3}{V^{3/2}\sqrt{n}}. \tag{110}$$

Now, define

$$E_n := \sqrt{n}[\mathsf{R}_0(r_{1,n}, r_{2,n}, D_1, D_2|Q_{XY}^{\hat{\mathbf{\Sigma}}_n}) - R_0^*] - \mathsf{A}_n + \lambda_1^* L_1 + \lambda_2^* L_2, \tag{111}$$

$$\ell := L_0 + \lambda_1^* L_1 + \lambda_2^* L_2. \tag{112}$$

Fix any positive real number $\zeta \in \mathbb{R}_+$. For sufficiently large n, it follows from (98) that

$$\varepsilon_n \le P_{XY}^n\{\mathsf{A}_n + E_n > \ell + b_n\} + P_{XY}^n\{\hat{\mathbf{\Sigma}}_n \notin \mathcal{K}_0\}, \tag{113}$$

$$\le P_{XY}^n\{\mathsf{A}_n > \ell - \zeta\} + P_{XY}^n\{|E_n| > \zeta\} + P_{XY}^n\{\hat{\mathbf{\Sigma}}_n \notin \mathcal{K}_0\}, \tag{114}$$

$$\le \mathrm{Q}\left(\frac{\ell-\zeta}{\sqrt{V}}\right) + \frac{C_{\mathrm{BE}}\beta_3}{V^{3/2}\sqrt{n}} + P_{XY}^n\{|E_n| > \zeta\} + P_{XY}^n\{\hat{\mathbf{\Sigma}}_n \notin \mathcal{K}_0\}, \tag{115}$$

where Eq. (114) follows from the union bound and the fact that $b_n \ge 0$, and Eq. (115) follows from (110). It follows from (100). that the last term in (115) vanishes. Furthermore, in Appendix F, we show that for any positive real number $\zeta \in \mathbb{R}_{+,0}$,

$$P_{XY}^n\{|E_n| > \zeta\} \longrightarrow 0. \tag{116}$$

Combining the above analyses, we have

$$\limsup_{n\to\infty} \varepsilon_n \le \mathrm{Q}\left(\frac{\ell-\zeta}{\sqrt{V}}\right). \tag{117}$$

Finally, by letting $\eta \to 0$, it follows from the definition of ℓ that

$$\limsup_{n\to\infty} \varepsilon_n \le \mathrm{Q}\left(\frac{L_0 + \lambda_1^* L_1 + \lambda_2^* L_2}{\sqrt{V}}\right). \tag{118}$$

The achievability proof of Theorem 1 is completed by using (48) and (97).

V. Converse Proof

We prove the converse part of Theorem 1 for an arbitrary sequence of codes satisfying (35) and (36). When $\lambda_1^* > 0$ and $\lambda_2^* > 0$, we use the converse bound of [12, Theorem 17] and specify how the entropy support formula of [12] leads to the desired converse bound. When either $\lambda_1^* = 0$ or $\lambda_2^* = 0$, we show how the problem reduces to the P2P case and uses the converse bound [6, Theorem 40].

For every fixed $\gamma > 0$ and all sufficiently large n, we consider any (n, M_0, M_1, M_2)-code per Def. 1 such that

$$\log M_{j,n} \le nR_j^* + \sqrt{n}(L_j + \gamma), \qquad j \in [0:2]. \tag{119}$$

A. Positive Supporting Weights

Suppose first that $\boldsymbol{\lambda}^* = (\lambda_1^*, \lambda_2^*) \in (0,1]^2$. We explain the covariance optimization used to relate the published converse to the selected boundary tuple. For a nearby covariance $\mathbf{\Sigma} \in \mathcal{K}$, the determinant characterization proved in Appendix E and (173) gives

$$\mathsf{R}_0(r_1, r_2, D_1, D_2 \mid Q_{XY}^{\mathbf{\Sigma}}) = \inf_{\substack{\mathbf{0}_{2\times 2} \prec \mathbf{\Sigma}_{\mathrm{N}} \preceq \mathbf{\Sigma} \\ \Sigma_{\mathrm{N},jj} \le D_j e^{2r_j},\ j\in[2]}} \frac{1}{2}\log\frac{|\mathbf{\Sigma}|}{|\mathbf{\Sigma}_{\mathrm{N}}|}, \qquad r_1, r_2 \ge 0. \tag{120}$$

For a fixed $\mathbf{\Sigma}_{\mathrm{N}}$, the Gaussian decomposition $(X,Y)^{\mathsf{T}} = \mathbf{W} + \mathbf{N}$, with independent covariances $\mathbf{\Sigma} - \mathbf{\Sigma}_{\mathrm{N}}$ and $\mathbf{\Sigma}_{\mathrm{N}}$, has common mutual information $\frac{1}{2}\log(|\mathbf{\Sigma}|/|\mathbf{\Sigma}_{\mathrm{N}}|)$. Its minimum private rates are $\frac{1}{2}[\log(\Sigma_{\mathrm{N},jj}/D_j)]^+$ for $j \in [2]$. Equivalently,

$$r_j \ge 0, \quad \Sigma_{\mathrm{N},jj} \le D_j e^{2r_j} \quad \Longleftrightarrow \quad r_j \ge \frac{1}{2}\left[\log\frac{\Sigma_{\mathrm{N},jj}}{D_j}\right]^+. \tag{121}$$

Thus, after fixing the residual covariance in (120), minimizing each private budget gives the right side of (121). Interchanging the two infima over this same feasible set yields

$$\inf_{\mathbf{r}\in\mathcal{R}(D_1,D_2|Q_{XY}^{\mathbf{\Sigma}})}(r_0+\lambda_1^*r_1+\lambda_2^*r_2)=\inf_{r_1,r_2\geq 0}\left[\mathsf{R}_0(r_1,r_2,D_1,D_2\mid Q_{XY}^{\mathbf{\Sigma}})+\lambda_1^*r_1+\lambda_2^*r_2\right] \tag{122}$$

$$=\inf_{\mathbf{0}_{2\times 2}\prec\mathbf{\Sigma}_{\mathrm{N}}\preceq\mathbf{\Sigma}}\left\{\frac{1}{2}\log\frac{|\mathbf{\Sigma}|}{|\mathbf{\Sigma}_{\mathrm{N}}|}+\sum_{j\in[2]}\frac{\lambda_j^*}{2}\left[\log\frac{\Sigma_{\mathrm{N},jj}}{D_j}\right]^+\right\}. \tag{123}$$

The first equality minimizes the common rate at fixed private rates. The second now follows explicitly from (120)–(121), since both weights are nonnegative.

The entropy expression in Liu et al. [12, Eq. (190)] has no positive parts. For $\mathbf{\Sigma}\in\mathbb{S}_{++}^2$, define

$$\underline{F}_{\boldsymbol{\lambda}^*}(\mathbf{\Sigma}):=\inf_{\mathbf{0}_{2\times 2}\prec\mathbf{\Sigma}_{\mathrm{N}}\preceq\mathbf{\Sigma}}\left\{\frac{1}{2}\log\frac{|\mathbf{\Sigma}|}{|\mathbf{\Sigma}_{\mathrm{N}}|}+\sum_{j\in[2]}\frac{\lambda_j^*}{2}\log\frac{\Sigma_{\mathrm{N},jj}}{D_j}\right\}. \tag{124}$$

For nearby covariances, this value is at most the operational support in (123). We next specify the covariance functional whose derivative produces the fluctuation in the published converse. Define

$$\varphi_{\boldsymbol{\lambda}^*,\mathbf{\Sigma}_{XY}}(\mathbf{\Sigma}):=-\underline{F}_{\boldsymbol{\lambda}^*}(\mathbf{\Sigma})+\frac{1}{2}\log|\mathbf{\Sigma}|-\frac{1}{2}\operatorname{tr}(\mathbf{\Sigma}_{XY}^{-1}\mathbf{\Sigma}). \tag{125}$$

This function is concave by [12, Proposition 5, Eqs. (64)–(74)]. A matrix $\mathbf{g}_{\mathrm{ent},\boldsymbol{\lambda}^*}\in\mathbb{S}^2$ is a supergradient at $\mathbf{\Sigma}_{XY}$ if for all $\mathbf{\Sigma}\in\mathbb{S}_{++}^2$,

$$\varphi_{\boldsymbol{\lambda}^*,\mathbf{\Sigma}_{XY}}(\mathbf{\Sigma})\leq\varphi_{\boldsymbol{\lambda}^*,\mathbf{\Sigma}_{XY}}(\mathbf{\Sigma}_{XY})+\langle\mathbf{g}_{\mathrm{ent},\boldsymbol{\lambda}^*},\mathbf{\Sigma}-\mathbf{\Sigma}_{XY}\rangle. \tag{126}$$

For $\mathbf{x}=(x,y)^{\mathsf{T}}$, define its centered quadratic fluctuation by

$$g_{\boldsymbol{\lambda}^*}(\mathbf{x}):=\mathbf{x}^{\mathsf{T}}\mathbf{g}_{\mathrm{ent},\boldsymbol{\lambda}^*}\mathbf{x}-\operatorname{tr}(\mathbf{g}_{\mathrm{ent},\boldsymbol{\lambda}^*}\mathbf{\Sigma}_{XY}). \tag{127}$$

The next lemma identifies the supporting value and this fluctuation in terms of quantities already used in the achievability proof.

Lemma 8. *For the selected boundary tuple with positive supporting weights, the following claims hold.*

(i) The value of the entropy support function satisfies

$$\underline{F}_{\boldsymbol{\lambda}^*}(\mathbf{\Sigma}_{XY})=R_0^*+\lambda_1^*R_1^*+\lambda_2^*R_2^*. \tag{128}$$

(ii) The entropy support is differentiable at the source covariance and

$$\nabla_{\mathbf{\Sigma}}\underline{F}_{\boldsymbol{\lambda}^*}(\mathbf{\Sigma})|_{\mathbf{\Sigma}=\mathbf{\Sigma}_{XY}}=\mathbf{g}. \tag{129}$$

(iii) The unique supergradient in (126) *is* $\mathbf{g}_{\mathrm{ent},\boldsymbol{\lambda}^*}=-\mathbf{g}$*, and*

$$g_{\boldsymbol{\lambda}^*}(\mathbf{x})=-[\jmath(x,y)-R_0^*], \tag{130}$$

$$\mathsf{Var}_{P_{XY}}[g_{\boldsymbol{\lambda}^*}(\mathbf{X})]=V. \tag{131}$$

The proof is provided in Appendix G. It establishes equality of the entropy and operational supports at the selected covariance, including zero private rates, and then identifies their touching derivative. Equality on a whole neighborhood is not needed.

We now apply the positive-weight Gaussian converse of Liu et al. [12, Theorem 17] to the identified entropy support and centered fluctuation.

In the notation of Liu et al., take $m=2$, $Q_{Y^m}=P_{XY}$, $c_0=1$, $c_j=\lambda_j^*$ for $j\in[2]$, and Lebesgue reference measures. Their entropy supporting value is

$$d^*:=\sup_{P_{U|XY}}\{-I(X,Y;U)-\lambda_1^*h(X\mid U)-\lambda_2^*h(Y\mid U)\}. \tag{132}$$

The supremum is over conditional distributions for which the displayed mutual information and conditional entropies are finite under $P_{XY}P_{U|XY}$. Finite auxiliary alphabets already suffice for this supporting value [12, Definition 3 and Remark 2]. Gaussian optimality [12, Proposition 5, Proposition 16, and Eq. (190)] and (124) give

$$d^*+\sum_{j\in[2]}\frac{\lambda_j^*}{2}\log(2\pi eD_j)=-\underline{F}_{\boldsymbol{\lambda}^*}(\mathbf{\Sigma}_{XY}). \tag{133}$$

Indeed, for a Gaussian common auxiliary with conditional covariance $\mathbf{\Sigma}_{\mathrm{N}}$, its conditional marginal entropies are $\frac{1}{2}\log(2\pi e\Sigma_{\mathrm{N},jj})$. Substituting these entropies into (132) gives (133). The Gaussian auxiliary in Lemma 8 attains this value. The source covariance is positive definite, both distortion levels and weights are positive, and the support is finite. Theorem 17 allows weights equal to one.

The functional subgradient in that theorem is normalized to have mean d^* [12, Eq. (52)]. Proposition 5 of that paper and Lemma 8 identify it as $d^* + g_{\boldsymbol{\lambda}^*}(\mathbf{x})$. Write

$$v_{j,n} := \mathrm{vol}\big(\mathcal{B}_n(\mathbf{0}_n, \sqrt{nD_j})\big), \qquad j \in [2], \tag{134}$$

where vol denotes Lebesgue volume. The finite-threshold form in the proof of [12, Theorem 17, Eq. (193)] states that for every $\theta \in \mathbb{R}$,

$$\varepsilon_n \geq P_{XY}^n \left\{ nd^* + \sum_{i\in[n]} g_{\boldsymbol{\lambda}^*}(\mathbf{X}_i) \leq \theta \right\} \tag{135}$$

$$- \exp\left\{ \theta + \log M_{0,n} + \sum_{j\in[2]} \lambda_j^* \log M_{j,n} + \sum_{j\in[2]} \lambda_j^* \log v_{j,n} \right\}. \tag{136}$$

Thus Liu's result supplies an excess-distortion lower bound for every code; Lemma 8 identifies exactly the supporting value and random fluctuation appearing in that bound.

The ball-volume formula and Stirling's expansion give

$$\log v_{j,n} = \log \frac{\pi^{n/2}(nD_j)^{n/2}}{\Gamma(n/2+1)} \tag{137}$$

$$= \frac{n}{2}\log(2\pi e D_j) - \frac{1}{2}\log(\pi n) + O(n^{-1}). \tag{138}$$

This is the estimate in [12, Eqs. (195)–(196)]. Taking $\theta = nd^* - \sqrt{n}a$ in (136) and using (133)–(138), we obtain, for every $a \in \mathbb{R}$ and sufficiently large n,

$$\varepsilon_n \geq P_{XY}^n \left\{ \frac{1}{\sqrt{n}} \sum_{i\in[n]} g_{\boldsymbol{\lambda}^*}(\mathbf{X}_i) \leq -a \right\} \tag{139}$$

$$- \exp\left\{ \log M_{0,n} + \sum_{j\in[2]} \lambda_j^* \log M_{j,n} - n\underline{F}_{\boldsymbol{\lambda}^*}(\boldsymbol{\Sigma}_{XY}) - \sqrt{n}a + C\log n \right\}, \tag{140}$$

where C is finite and depends only on the fixed distortions and weights. By Lemma 8, the summands in the probability are independent and centered with variance V. Their third absolute moment equals the finite β_3 in (110). The Berry–Esseen bound therefore gives

$$P_{XY}^n \left\{ \frac{1}{\sqrt{n}} \sum_{i\in[n]} g_{\boldsymbol{\lambda}^*}(\mathbf{X}_i) \leq -a \right\} \geq \mathrm{Q}(a/\sqrt{V}) - \frac{C_{\mathrm{BE}}\beta_3}{V^{3/2}\sqrt{n}}. \tag{141}$$

Define $\ell := L_0 + \lambda_1^* L_1 + \lambda_2^* L_2$. Fix $\gamma > 0$ and take $a_\gamma := \ell + (2 + \lambda_1^* + \lambda_2^*)\gamma$. From (119) and Lemma 8, for sufficiently large n,

$$\log M_{0,n} + \sum_{j\in[2]} \lambda_j^* \log M_{j,n} - n\underline{F}_{\boldsymbol{\lambda}^*}(\boldsymbol{\Sigma}_{XY}) - \sqrt{n}a_\gamma \leq \sqrt{n}\big[\ell + (1 + \lambda_1^* + \lambda_2^*)\gamma - a_\gamma\big] \tag{142}$$

$$= -\gamma\sqrt{n}. \tag{143}$$

Combining (140)–(143) yields

$$\varepsilon_n \geq \mathrm{Q}(a_\gamma/\sqrt{V}) - \frac{C_{\mathrm{BE}}\beta_3}{V^{3/2}\sqrt{n}} - \exp\{-\gamma\sqrt{n} + C\log n\}. \tag{144}$$

Taking the limit inferior as $n \to \infty$ and then letting $\gamma \downarrow 0$ gives the desired converse directly:

$$\liminf_{n\to\infty} \varepsilon_n \geq \mathrm{Q}\left(\frac{L_0 + \lambda_1^* L_1 + \lambda_2^* L_2}{\sqrt{V}} \right). \tag{145}$$

The argument uses only the one-sided rate envelopes and also covers normalized weighted displacements tending to minus infinity.

For the original code sequence, (36) and (145) imply

$$L_0 + \lambda_1^* L_1 + \lambda_2^* L_2 \geq \sqrt{V}\,\mathrm{Q}^{-1}(\varepsilon), \tag{146}$$

since $\liminf_n \varepsilon_n \leq \limsup_n \varepsilon_n \leq \varepsilon$ and Q is decreasing. This proves the required converse inequality whenever both weights are positive.

B. A Zero Supporting Weight

By Lemma 3 and the branch derivatives in Appendix B, a zero supporting weight at a positive-common-rate boundary tuple has either $(\lambda_1^*, \lambda_2^*) = (1, 0)$ or $(0, 1)$. In the first case,

$$R_0^* + R_1^* = R_X(P_X, D_1), \qquad V = 1/2. \tag{147}$$

Decoder 1 has at most $M_{0,n}M_{1,n}$ reconstruction vectors. Enumerate these vectors in a fixed order. For every x^n, choose the smallest index minimizing the distortion to x^n. This defines a Borel encoder observing only X^n, because it selects a minimum among finitely many continuous functions with a fixed tie rule. Whenever the original decoder 1 reconstructs within distortion D_1, this scalar encoder also does so. Its excess-distortion probability is therefore at most the original joint probability ε_n, even though the original encoder also observes Y^n.

Fix $0 < \delta < 1 - \varepsilon$. By (36), the aggregate scalar code has excess-distortion probability at most $\varepsilon + \delta$ for all sufficiently large n. The scalar Gaussian converse [6, Theorem 40], applied at this fixed error level, gives

$$\log(M_{0,n}M_{1,n}) \geq nR_X(P_X, D_1) + \sqrt{n/2}\,\mathrm{Q}^{-1}(\varepsilon + \delta) - C_\delta \log n \tag{148}$$

for some finite C_δ and sufficiently large n. Meanwhile, (119) gives

$$\frac{\log(M_{0,n}M_{1,n}) - nR_X(P_X, D_1)}{\sqrt{n}} = \frac{\log M_{0,n} - nR_0^* + \log M_{1,n} - nR_1^*}{\sqrt{n}} \tag{149}$$

$$\leq L_0 + L_1 + 2\gamma. \tag{150}$$

Combining (148)–(150), letting $n \to \infty$, and then letting $\gamma \downarrow 0$ and $\delta \downarrow 0$ yields

$$L_0 + \lambda_1^* L_1 + \lambda_2^* L_2 = L_0 + L_1 \geq \frac{\mathrm{Q}^{-1}(\varepsilon)}{\sqrt{2}} = \sqrt{V}\,\mathrm{Q}^{-1}(\varepsilon). \tag{151}$$

When $(\lambda_1^*, \lambda_2^*) = (0, 1)$, apply the same construction to decoder 2's $M_{0,n}M_{2,n}$ reconstruction vectors and the source Y^n. The identities $R_0^* + R_2^* = R_Y(P_Y, D_2)$ and $V = 1/2$ give the same inequality with $L_0 + L_2$ on the left. These two arguments include the zero-weight interfaces.

Thus every admissible code sequence satisfies the supporting-half-space inequality and $L_j \geq 0$ whenever $R_j^* = 0$. This half-space intersected with the private-coordinate constraints is closed, so taking the closure in Definition 4 preserves the inclusion. Together with the achievability proof, this completes the proof of Theorem 1.

VI. Conclusion

In this paper, we derived exact second-order asymptotics for the lossy Gaussian Gray–Wyner problem at boundary rate tuples with a nonzero common rate under the quadratic distortion measure and a joint excess-distortion probability criterion. Our main contribution was an achievability bound that matches the existing converse bounds [6], [12]. We obtained this bound by deriving a covariance shell covering lemma analogous to the type covering lemma for discrete sources.

We next discuss future directions. Firstly, in this paper, we derived the second-order asymptotics. However, to accurately characterize the performance of optimal codes at small blocklengths, further refined asymptotics including the third-order asymptotics [6, Theorem 27] or the refined moderate deviations [31] are necessary. Secondly, we considered boundary rate tuples with a nonzero common rate in this paper. It is possible that some boundary rate tuple has a zero common rate, which can still have a nonzero second-order rate. It would be worthwhile to study the corresponding second-order coding for the Gray-Wyner problem in this regime. Finally, one can generalize our second-order asymptotics to other multiterminal lossy source coding problems concerning bivariate Gaussian sources, e.g., the Wyner–Ziv problem [32] or the Heegard–Berger problem [33]. The techniques in this paper, especially covariance shell covering could be helpful.

Declaration of AI Usage

The authors formulated the Gaussian Gray-Wyner problem, pointed out the research gap and provided proof strategies. ChatGPT (Astra 6 Ultra mode) was used to test proof strategies, fill proof steps and prepare drafts. The authors have reviewed AI-assisted derivations and assume full responsibility for the statements, proofs, and correctness of the mathematical content. In particular, the authors have rewritten the main text and revised most contents in the appendix.

APPENDIX A
PROOF OF LEMMA 3

By (29), a boundary tuple must satisfy $R_1^* \le R_X(P_X, D_1)$, $R_2^* \le R_Y(P_Y, D_2)$, and $R_0^* = \Upsilon_{\rho_{XY}}(\alpha^*, \beta^*)$. Both deficits are positive when $R_0^* > 0$: if $\alpha^* > 0$ and $\beta^* = 0$, then $R_2^* = R_Y(P_Y, D_2) > 0$. For sufficiently small $t \in (0, R_2^*]$ with $\nu_t < \rho_{XY}\nu_{\alpha^*}$, reducing R_2^* by t leaves the minimum common rate equal to α^*, contradicting Pareto minimality. The other axis is symmetric; at the origin the common rate vanishes.

For $(\alpha^*, \beta^*) \in \mathcal{P}_1 \cup \mathcal{P}_2$, the two diagonal caps are active in the unique optimizer (171). Reducing either positive private rate strictly tightens its cap and excludes this optimizer. If the maximum determinant remained unchanged, an optimizer for the tightened feasible set would also maximize the old determinant, contradicting uniqueness proved in Appendix E. Thus every tuple on these parts of the lower-boundary graph is Pareto-minimal, including the interfaces. Substitution of (27) and (32) gives

$$\begin{aligned} R_0^* + R_1^* + R_2^* &= R_X(P_X, D_1) + R_Y(P_Y, D_2) + \Upsilon_{\rho_{XY}}(\alpha^*, \beta^*) - \alpha^* - \beta^* \\ &= \begin{cases} R_X(P_X, D_1) + R_Y(P_Y, D_2) - \dfrac{1}{2}\log\dfrac{1 - r^2_{\alpha^*,\beta^*}}{1 - \rho_{XY}^2}, & \text{if } (\alpha^*, \beta^*) \in \mathcal{P}_1, \\ R_X(P_X, D_1) + R_Y(P_Y, D_2) + \dfrac{1}{2}\log(1 - \rho_{XY}^2), & \text{if } (\alpha^*, \beta^*) \in \mathcal{P}_2. \end{cases} \end{aligned}$$

In the strict $\mathcal{P}_3$ regime, $\Upsilon_{\rho_{XY}}(\alpha^*, \beta^*) = \alpha^*$. If $R_2^* > 0$, a sufficiently small reduction of this private rate stays in $\mathcal{P}_3$ and does not change the common rate, so the tuple is not Pareto-minimal. If $R_2^* = 0$, this coordinate cannot be reduced; reducing a positive R_1^* increases α^* and therefore the common-rate lower bound $\Upsilon_{\rho_{XY}} \ge \alpha$. The boundary relation is consequently

$$R_0^* + R_1^* = R_X(P_X, D_1).$$

The symmetric argument gives $R_1^* = 0$ and $R_0^* + R_2^* = R_Y(P_Y, D_2)$ in the strict $\mathcal{P}_4$ regime. Interfaces are already included in $\mathcal{P}_1 \cup \mathcal{P}_2$. These regimes exhaust all candidates with positive common rate, proving the lemma.

APPENDIX B
PROOF OF PROPOSITION 1

Proof of Statement (i). For $\alpha, \beta > 0$, write $r = r_{\alpha,\beta}$ and $u = (\nu_\beta/\nu_\alpha)e^{\beta-\alpha}$. Since $\nu_t' = e^{-2t}/\nu_t$, differentiation of (22) gives $\partial_\alpha r = r - u$, $\partial_\beta r = r - u^{-1}$, and $\partial_\rho r = e^{\alpha+\beta}$. Let F_k denote expression k of (27). The curved expression has derivatives

$$\partial_\alpha F_1 = 1 + \frac{r(r-u)}{1-r^2} = \frac{1-ru}{1-r^2}, \tag{152}$$

$$\partial_\beta F_1 = 1 + \frac{r(r-u^{-1})}{1-r^2} = \frac{1-r/u}{1-r^2}, \tag{153}$$

$$\partial_\rho F_1 = \frac{re^{\alpha+\beta}}{1-r^2} - \frac{\rho}{1-\rho^2}. \tag{154}$$

On the other three regimes,

$$(\partial_\alpha F_k, \partial_\beta F_k, \partial_\rho F_k) = \begin{cases} (1, 1, -\rho/(1-\rho^2)), & k = 2, \\ (1, 0, 0), & k = 3, \\ (0, 1, 0), & k = 4. \end{cases} \tag{155}$$

The curved-regime constraints imply $0 \le r \le \min\{u, u^{-1}\} < 1$ except possibly when $u = 1$. In that case $\alpha = \beta$, and $r = 1 - (1-\rho)e^{2\alpha} < 1$. Thus the curved expression is analytic on an open neighborhood of each of its finite interfaces.

At the $(\mathcal{P}_1, \mathcal{P}_2)$ interface, $r = 0$ and (152)–(154) give $\nabla F_1 = (1, 1, -\rho/(1-\rho^2)) = \nabla F_2$. At the $(\mathcal{P}_1, \mathcal{P}_3)$ interface, $\nu_\beta = \rho\nu_\alpha$, $r = u$, and

$$\begin{aligned} 1 - r^2 &= (1-\rho^2)e^{2\beta}, \\ \frac{re^{\alpha+\beta}}{1-r^2} &= \frac{\rho}{1-\rho^2}. \end{aligned}$$

Hence $F_1 = \alpha = F_3$ and $\nabla F_1 = (1, 0, 0) = \nabla F_3$. Interchanging the two deficits proves the same agreement at the $(\mathcal{P}_1, \mathcal{P}_4)$ interface. The values and first derivatives of the adjoining analytic expressions therefore agree continuously. This proves Statement (i).

Proof of Statement (ii). Lemma 3 gives $\alpha^*, \beta^* > 0$. Hence an open neighborhood $\mathcal{U}$ of $(\mathbf{\Sigma}_{XY}, R_1^*, R_2^*)$ can be chosen where the covariance is positive definite, $\Sigma_{jj} > D_j$, $\Sigma_{12} > 0$, and both logarithmic marginal deficits remain positive. On $\mathcal{U}$, the map

$$(\mathbf{\Sigma}, r_1, r_2) \mapsto \left(\frac{\Sigma_{12}}{\sqrt{\Sigma_{11}\Sigma_{22}}}, \frac{1}{2}\log\frac{\Sigma_{11}}{D_1} - r_1, \frac{1}{2}\log\frac{\Sigma_{22}}{D_2} - r_2 \right)$$

is smooth. Composing it with Υ yields a C^1 extension of the operational common-rate function to $\mathcal{U}$, including small negative private rates used only to define the derivatives at zero.

For a symmetric perturbation $\mathbf{h} = [h_{jk}]$ and private-rate perturbations (s_1, s_2), write $(\eta, \lambda_1, \lambda_2)$ for the derivatives of Υ at the local covariance and deficits. Direct differentiation gives

$$\begin{aligned}\mathrm{d}\overline{R}_0[\mathbf{h}, s_1, s_2] &= \lambda_1 \left(\frac{h_{11}}{2\Sigma_{11}} - s_1 \right) + \lambda_2 \left(\frac{h_{22}}{2\Sigma_{22}} - s_2 \right) \\ &\quad + \eta \left(\frac{h_{12}}{\sqrt{\Sigma_{11}\Sigma_{22}}} - \frac{\rho(\mathbf{\Sigma}) h_{11}}{2\Sigma_{11}} - \frac{\rho(\mathbf{\Sigma}) h_{22}}{2\Sigma_{22}} \right) \\ &= \frac{\lambda_1 - \rho(\mathbf{\Sigma})\eta}{2\Sigma_{11}} h_{11} + \frac{\eta}{\sqrt{\Sigma_{11}\Sigma_{22}}} h_{12} + \frac{\lambda_2 - \rho(\mathbf{\Sigma})\eta}{2\Sigma_{22}} h_{22} - \lambda_1 s_1 - \lambda_2 s_2. \end{aligned} \tag{156}$$

The coefficients are continuous on $\mathcal{U}$. At $\mathbf{\Sigma} = \mathbf{\Sigma}_{XY}$, the identity $\langle \mathbf{g}, \mathbf{h} \rangle = g_{11}h_{11} + 2g_{12}h_{12} + g_{22}h_{22}$ identifies (40) and the private derivatives $-\lambda_j^*$.

For completeness, the zero-slope interface mentioned in Section III-A has a quadratic increase in the suppressed direction. At $\nu_\beta = \rho\nu_\alpha$, the first β-derivative vanishes, whereas differentiating (153) on the $\mathcal{P}_1$ side yields

$$\begin{aligned}\partial_{\beta,+}^2 \Upsilon_\rho(\alpha, \beta) &= u^{-2} + \frac{1}{\nu_\beta^2 (1 - u^2)} \\ &= \frac{(1 - \rho^2\nu_\alpha^2)^2}{\rho^2 \nu_\alpha^2 (1 - \nu_\alpha^2)(1 - \rho^2)} > 0. \end{aligned}$$

Indeed, $\partial_\beta r = r - u^{-1}$, $\partial_\beta \log u = \nu_\beta^{-2}$, and $u^2 = \rho^2(1 - \nu_\alpha^2)/(1 - \rho^2\nu_\alpha^2)$ at this interface. Taylor's formula therefore gives

$$\Upsilon_\rho(\alpha, \beta + h) = \alpha + \frac{(1 - \rho^2\nu_\alpha^2)^2}{2\rho^2 \nu_\alpha^2 (1 - \nu_\alpha^2)(1 - \rho^2)} h^2 + o(h^2), \quad h \downarrow 0. \tag{157}$$

For small $h < 0$, the value remains α in $\mathcal{P}_3$. This completes Statement (ii).

Appendix C
Proof of Lemma 4

For $\mathbf{X} = (X, Y)^\mathsf{T} \sim \mathcal{N}(\mathbf{0}_2, \mathbf{\Sigma})$, the definition (42) and $\mathsf{E}[\mathbf{X}\mathbf{X}^\mathsf{T}] = \mathbf{\Sigma}$ give

$$\mathsf{E}_{Q_{XY}^{\mathbf{\Sigma}}}[\jmath(X, Y)] = R_0^* + \operatorname{tr}[\mathbf{g}(\mathbf{\Sigma} - \mathbf{\Sigma}_{XY})].$$

Setting $\mathbf{\Sigma} = \mathbf{\Sigma}_{XY}$ proves claim (i).

Let $\boldsymbol{\omega} := \mathbf{\Sigma}_{XY}^{1/2} \mathbf{g} \mathbf{\Sigma}_{XY}^{1/2}$ have eigenvalues ω_1, ω_2. Whitening $\mathbf{X}$ and orthogonally diagonalizing $\boldsymbol{\omega}$ yield independent standard Gaussians N_1, N_2 for which

$$\jmath(X, Y) - R_0^* \overset{\mathrm{d}}{=} \omega_1(N_1^2 - 1) + \omega_2(N_2^2 - 1), \tag{158}$$

$$V = 2(\omega_1^2 + \omega_2^2) = 2 \operatorname{tr}[(\mathbf{g}\mathbf{\Sigma}_{XY})^2]. \tag{159}$$

The variance identity uses $\mathsf{Var}(N_j^2) = 2$ and independence. This representation also gives a finite third absolute moment.

It remains to identify the eigenvalues. Let $\mathbf{\Sigma}_{\mathrm{N}}^*$ be the unique optimizer in (67) at the selected tuple, and put $\tau_j = D_j e^{2R_j^*}$. The determinant maximization is equivalent, up to a covariance-dependent constant, to minimizing $-\frac{1}{2}\log|\mathbf{\Sigma}_{\mathrm{N}}|$ subject to $\mathbf{\Sigma}_{\mathrm{N}} \preceq \mathbf{\Sigma}_{XY}$ and $\Sigma_{\mathrm{N},jj} \le \tau_j$. A sufficiently small $\delta\mathbf{1}_2$ is strictly feasible. Slater's condition therefore gives multipliers $\vartheta_j \ge 0$ and $\boldsymbol{\lambda}_{\mathrm{cov}} \succeq 0$ satisfying

$$-\frac{1}{2}(\mathbf{\Sigma}_{\mathrm{N}}^*)^{-1} + \operatorname{diag}(\vartheta_1, \vartheta_2) + \boldsymbol{\lambda}_{\mathrm{cov}} = \mathbf{0}_{2\times 2}, \tag{160}$$

$$\langle \boldsymbol{\lambda}_{\mathrm{cov}}, \mathbf{\Sigma}_{XY} - \mathbf{\Sigma}_{\mathrm{N}}^* \rangle = 0, \tag{161}$$

together with $\vartheta_j(\Sigma_{\mathrm{N},jj}^* - \tau_j) = 0$. Keeping the multipliers and caps fixed while perturbing the source covariance, weak duality and equality at $\mathbf{\Sigma}_{XY}$ give

$$\mathsf{R}_0(R_1^*, R_2^*, D_1, D_2 \mid Q_{XY}^{\mathbf{\Sigma}}) - \frac{1}{2}\log|\mathbf{\Sigma}| \ge R_0^* - \frac{1}{2}\log|\mathbf{\Sigma}_{XY}| - \langle \boldsymbol{\lambda}_{\mathrm{cov}}, \mathbf{\Sigma} - \mathbf{\Sigma}_{XY} \rangle. \tag{162}$$

Indeed, by (173), the left-hand side is the minimum of $-\frac{1}{2}\log|\mathbf{\Sigma}_{\mathrm{N}}|$ under the perturbed covariance constraint. The two sides touch at $\mathbf{\Sigma}_{XY}$. Differentiating there in both signs of every symmetric direction, using Proposition 1, yields

$$\mathbf{g} = \frac{1}{2}\mathbf{\Sigma}_{XY}^{-1} - \boldsymbol{\lambda}_{\mathrm{cov}}. \tag{163}$$

Set $\mathbf{\Sigma}_{\mathrm{W}}^{*} := \mathbf{\Sigma}_{XY} - \mathbf{\Sigma}_{\mathrm{N}}^{*} \succeq 0$. Since $R_0^{*} > 0$, this matrix is nonzero. Positive semidefiniteness and (161) imply

$$0 = \operatorname{tr}(\boldsymbol{\lambda}_{\mathrm{cov}} \mathbf{\Sigma}_{\mathrm{W}}^{*}) = \left\| \boldsymbol{\lambda}_{\mathrm{cov}}^{1/2} (\mathbf{\Sigma}_{\mathrm{W}}^{*})^{1/2} \right\|^2 .$$

Consequently $\boldsymbol{\lambda}_{\mathrm{cov}} \mathbf{\Sigma}_{\mathrm{W}}^{*} = \mathbf{0}_{2\times 2}$, so the nonzero range of $\mathbf{\Sigma}_{\mathrm{W}}^{*}$ lies in the kernel of $\boldsymbol{\lambda}_{\mathrm{cov}}$. In dimension two this implies $\operatorname{rank}(\boldsymbol{\lambda}_{\mathrm{cov}}) \le 1$. Equation (163) then gives

$$\boldsymbol{\omega} = \frac{1}{2}\mathbf{1}_2 - \mathbf{\Sigma}_{XY}^{1/2} \boldsymbol{\lambda}_{\mathrm{cov}} \mathbf{\Sigma}_{XY}^{1/2},$$

and hence one eigenvalue of $\boldsymbol{\omega}$ equals $1/2$. From (40),

$$\operatorname{tr}(\boldsymbol{\omega}) = \operatorname{tr}(\mathbf{g}\mathbf{\Sigma}_{XY}) = \frac{\lambda_1^{*} + \lambda_2^{*}}{2}.$$

Thus the other eigenvalue is $(\lambda_1^{*} + \lambda_2^{*} - 1)/2$. Substitution into (159) proves (46).

In $\mathcal{P}_1$, write $r = r_{\alpha^{*},\beta^{*}}$ and recall u from (44). The derivatives in (152)–(153), equivalently (187)–(188), give

$$\lambda_1^{*} + \lambda_2^{*} - 1 = \frac{1 + r^2 - r(u + u^{-1})}{1 - r^2} = \frac{(u - r)(1 - ru)}{u(1 - r^2)}. \tag{164}$$

In the strict curved regime $0 < r < \min\{u, u^{-1}\}$, so the last expression is positive, while each slope lies in $(0, 1)$. In $\mathcal{P}_2, \mathcal{P}_3, \mathcal{P}_4$ the slopes are respectively $(1, 1), (1, 0), (0, 1)$ by (155); the adjoining values agree by continuity. Therefore $1 \le \lambda_1^{*} + \lambda_2^{*} \le 2$, and substitution in (46) gives all four expressions in (47), including their interfaces, together with $1/2 \le V \le 1$. These arguments remain valid when a private rate is zero because its cap remains strictly positive and the covariance derivative exists by Proposition 1.

Appendix D
Proof of Lemma 5

We first prove a covering bound uniform in the ratio of the two radii. Fix $T \ge 1$, $n \ge 3$, and $0 \le R \le Tr$ with $r > 0$. We seek a finite $\mathcal{V} \subset \mathbb{R}^n$ such that

$$\mathcal{B}_n(\mathbf{0}_n, R) \subseteq \bigcup_{\mathbf{v}\in\mathcal{V}} \mathcal{B}_n(\mathbf{v}, r).$$

Its cardinality must satisfy

$$|\mathcal{V}| \le C_T n^4 \max\{1, (R/r)^n\}.$$

If $R \le r$, take $\mathcal{V} = \{\mathbf{0}_n\}$. Suppose $R > r$, and put $h = r/(4n)$, $r_{-} = r - 2h$, $m = \lceil (R - r)/h \rceil$, and $R_k = \min\{r + kh, R\}$ for $k \in [m]$. On each sphere $\mathcal{S}_k := \{\mathbf{x} : \|\mathbf{x}\| = R_k\}$, take a maximal h-separated set $\mathcal{N}_k$. It is an h-net by maximality. Packing radius-$h/2$ balls and comparing their volumes give

$$|\mathcal{N}_k| \le \left(1 + \frac{2R_k}{h}\right)^n \le (1 + 8Tn)^n. \tag{165}$$

For each k, choose a center uniformly on the sphere of radius $a_k = \sqrt{R_k^2 - r_{-}^2}$. A fixed net point $\mathbf{z}$ with $\|\mathbf{z}\| = R_k$ is covered by the radius-r_{-} ball about this center precisely when the angle θ satisfies $\cos\theta \ge a_k/R_k =: z_k$. The first coordinate of a uniform point on the unit sphere has density $\omega_n(1 - u^2)^{(n-3)/2}$, with $\omega_n := \Gamma(n/2)/(\sqrt{\pi}\,\Gamma((n-1)/2)) \ge 1/2$. Substitution $w = 1 - u^2$ and $(1 - w)^{-1/2} \ge 1$ show that the coverage probability satisfies

$$\begin{aligned} p_k &= \omega_n \int_{z_k}^{1} (1 - u^2)^{(n-3)/2}\, du \\ &= \frac{\omega_n}{2} \int_{0}^{r_{-}^2/R_k^2} w^{(n-3)/2} (1 - w)^{-1/2}\, dw \\ &\ge \frac{1}{2(n-1)} \left(\frac{r_{-}}{R_k}\right)^{n-1}. \end{aligned} \tag{166}$$

The inequality $\omega_n \ge 1/2$ follows by integrating its density over $[-1, 1]$, and the bound remains uniform when $R/r \downarrow 1$.

Independently draw

$$J_k := \left\lceil 2(n-1) \left(\frac{R_k}{r_{-}}\right)^{n-1} [n \log(1 + 8Tn) + 1] \right\rceil \tag{167}$$

centers for this sphere. Since $J_k p_k \ge n\log(1 + 8Tn) + 1$, the union bound and (165) imply that the probability any point of $\mathcal{N}_k$ is uncovered is at most

$$|\mathcal{N}_k|(1 - p_k)^{J_k} \le (1 + 8Tn)^n e^{-J_k p_k} \le e^{-1} < 1.$$

Fix successful center sets $\mathcal{C}_k$ for all k, and put $\mathcal{V} = \{\mathbf{0}_n\} \cup \bigcup_{k\in[m]} \mathcal{C}_k$. For $r < \|\mathbf{x}\| \le R$, choose R_k within h of $\|\mathbf{x}\|$, radially project $\mathbf{x}$ onto $\mathcal{S}_k$, and choose a net point within distance h. A center in $\mathcal{C}_k$ lies within distance r_- of that net point. The triangle inequality gives total distance at most $2h + r_- = r$. The origin covers the inner ball.

Finally, $m \le 4nT + 1$ and

$$\left(\frac{r}{r_-}\right)^{n-1} = \left(1 - \frac{1}{2n}\right)^{-(n-1)} \le e^{1/2}, \tag{168}$$

where $-\log(1-x) \le x/(1-x)$. Since $R_k \le R$, the ceiling bound in (167) yields

$$\begin{aligned} |\mathcal{V}| &\le 1 + \sum_{k\in[m]} J_k \\ &\le 1 + m + 2m(n-1)e^{1/2}(R/r)^{n-1}[n\log(1+8Tn)+1] \\ &\le C_T n^4 (R/r)^n, \end{aligned} \tag{169}$$

where C_T depends only on T, using $m \le 4nT+1$ and $1 + \log n \le n$ for $n \ge 3$.

To apply this bound to Lemma 5, choose

$$T := \max\left\{1, \sqrt{\frac{\kappa^{-1} + \chi_\kappa}{\min\{D_1, D_2\}}}\right\}.$$

By $s_n/n \le 1$ and $\mathbf{\Sigma}_{\mathrm{N},q} \preceq \kappa^{-1}\mathbf{1}_2$, we have $\xi_{j,q,n}^2/(nD_j) \le T^2$. Taking $R = \xi_{j,q,n}$ and $r = \sqrt{nD_j}$ gives

$$M_{j,q} \le C_T n^4 \max\left\{1, \left(\frac{\xi_{j,q,n}^2}{nD_j}\right)^{n/2}\right\}.$$

The ordered centers form deterministic private codebooks satisfying (73); $C_{\mathrm{cov}} = C_T$ depends only on (κ, D_1, D_2). This completes the proof.

Appendix E
Proof of Lemma 6

Fix $\mathbf{\Sigma} \in \mathcal{K}$. Since $\|\mathbf{\Sigma} - \mathbf{\Sigma}_{XY}\| \le 2\eta_{\mathrm{cov}}$, the eigenvalue and entry perturbation bounds in (55) give

$$\begin{aligned} \lambda_{\min}(\mathbf{\Sigma}) &\ge \lambda_{\min}(\mathbf{\Sigma}_{XY}) - 2\eta_{\mathrm{cov}} \ge \frac{3}{4}\lambda_{\min}(\mathbf{\Sigma}_{XY}) > 0, \\ \Sigma_{jj} - D_j &\ge [\mathbf{\Sigma}_{XY}]_{jj} - D_j - 2\eta_{\mathrm{cov}} \ge \frac{3}{4}([\mathbf{\Sigma}_{XY}]_{jj} - D_j) > 0, \qquad j \in [2]. \end{aligned}$$

The first line follows from the Rayleigh quotient and $\|A\|_{\mathrm{op}} \le \|A\|$; the second uses the entrywise Frobenius bound. Compactness yields $0 < m < M$ with $m\mathbf{1}_2 \preceq \mathbf{\Sigma} \preceq M\mathbf{1}_2$ throughout $\mathcal{K}$. The covariance segment from $\mathbf{\Sigma}_{XY}$ to $\mathbf{\Sigma}$ has diagonal entries at least half their source values. Differentiating (18) along this segment and applying the mean-value bound therefore gives

$$|\rho(\mathbf{\Sigma}) - \rho_{XY}| \le c_{\mathrm{cor}} \|\mathbf{\Sigma} - \mathbf{\Sigma}_{XY}\| \le 2c_{\mathrm{cor}}\eta_{\mathrm{cov}} \le \frac{1}{4}\min\{\rho_{XY}, 1-\rho_{XY}\}.$$

Hence $\rho_{XY}/2 \le \rho(\mathbf{\Sigma}) \le (1+\rho_{XY})/2$, so the Gaussian first-order formula applies uniformly on $\mathcal{K}$.

Choose $0 < \delta < \min\{m, D_1, D_2\}$ and $0 < \kappa < 1$ such that $\kappa \le \min\{\delta^2/M, 1/M\}$. For every nonnegative private rate pair, $\delta\mathbf{1}_2$ is feasible in (67): the inequalities $\delta < m$ and $\delta < D_j \le D_j e^{2r_j}$ verify the covariance order and both diagonal caps. The feasible set is closed and bounded in the finite-dimensional space $\mathbb{S}^2$, so continuity of the determinant gives an optimizer $\mathbf{\Sigma}_{\mathrm{N}}^*$ with

$$|\mathbf{\Sigma}_{\mathrm{N}}^*| \ge |\delta\mathbf{1}_2| = \delta^2 > 0.$$

The optimizer is positive semidefinite by feasibility and has positive determinant, so it is positive definite. Since $\lambda_{\max}(\mathbf{\Sigma}_{\mathrm{N}}^*) \le M$, the product of its two eigenvalues gives

$$\lambda_{\min}(\mathbf{\Sigma}_{\mathrm{N}}^*) = \frac{|\mathbf{\Sigma}_{\mathrm{N}}^*|}{\lambda_{\max}(\mathbf{\Sigma}_{\mathrm{N}}^*)} \ge \frac{\delta^2}{M} \ge \kappa.$$

The upper bound $\mathbf{\Sigma} \preceq \kappa^{-1}\mathbf{1}_2$ follows from $\kappa \le 1/M$. Combining these bounds gives

$$\kappa\mathbf{1}_2 \preceq \mathbf{\Sigma}_{\mathrm{N}}^* \preceq \mathbf{\Sigma} \preceq \kappa^{-1}\mathbf{1}_2. \tag{170}$$

The same κ applies throughout $\mathcal{K}$ and to all nonnegative private rates. In particular, setting $\mathbf{\Sigma} = \mathbf{\Sigma}_q$ gives these bounds for the optimizer $\mathbf{\Sigma}_{\mathrm{N},q}$ in $\mathcal{F}_q$.

We next verify the optimizer's value explicitly. Write $\rho_{\mathbf{\Sigma}} := \rho(\mathbf{\Sigma})$, $\alpha := \alpha(\mathbf{\Sigma}; r_1, D_1)$, and $\beta := \beta(\mathbf{\Sigma}; r_2, D_2)$. Define $\mathbf{t}_{\mathbf{\Sigma}} := \operatorname{diag}(\sqrt{\Sigma_{11}}, \sqrt{\Sigma_{22}})$ and

$$v_1 := e^{-2\alpha} = \min\{1, D_1 e^{2r_1}/\Sigma_{11}\}, \qquad v_2 := e^{-2\beta} = \min\{1, D_2 e^{2r_2}/\Sigma_{22}\}.$$

Then $\nu_\alpha = \sqrt{1 - v_1}$ and $\nu_\beta = \sqrt{1 - v_2}$. For positive deficits, consider the normalized covariance

$$\widetilde{\mathbf{\Sigma}}^*_{\mathrm{N}} := \begin{cases} \begin{bmatrix} v_1 & \rho_{\mathbf{\Sigma}} - \nu_\alpha \nu_\beta \\ \rho_{\mathbf{\Sigma}} - \nu_\alpha \nu_\beta & v_2 \end{bmatrix}, & \text{if } (\alpha, \beta) \in \mathcal{P}_1, \\ \begin{bmatrix} v_1 & 0 \\ 0 & v_2 \end{bmatrix}, & \text{if } (\alpha, \beta) \in \mathcal{P}_2, \\ \begin{bmatrix} v_1 & \rho_{\mathbf{\Sigma}} v_1 \\ \rho_{\mathbf{\Sigma}} v_1 & 1 - \rho^2_{\mathbf{\Sigma}} + \rho^2_{\mathbf{\Sigma}} v_1 \end{bmatrix}, & \text{if } (\alpha, \beta) \in \mathcal{P}_3, \\ \begin{bmatrix} 1 - \rho^2_{\mathbf{\Sigma}} + \rho^2_{\mathbf{\Sigma}} v_2 & \rho_{\mathbf{\Sigma}} v_2 \\ \rho_{\mathbf{\Sigma}} v_2 & v_2 \end{bmatrix}, & \text{if } (\alpha, \beta) \in \mathcal{P}_4. \end{cases} \tag{171}$$

Here the regime definitions are evaluated with correlation $\rho_{\mathbf{\Sigma}}$. Set $\mathbf{\Sigma}^*_{\mathrm{N}} := \mathbf{t}_{\mathbf{\Sigma}} \widetilde{\mathbf{\Sigma}}^*_{\mathrm{N}} \mathbf{t}_{\mathbf{\Sigma}}$ and $\widetilde{\mathbf{\Sigma}} := \mathbf{t}_{\mathbf{\Sigma}}^{-1} \mathbf{\Sigma} \mathbf{t}_{\mathbf{\Sigma}}^{-1}$. The latter matrix has unit diagonal and off-diagonal entry $\rho_{\mathbf{\Sigma}}$.

On $\mathcal{P}_1$, subtraction yields

$$\widetilde{\mathbf{\Sigma}} - \widetilde{\mathbf{\Sigma}}^*_{\mathrm{N}} = \begin{bmatrix} \nu_\alpha \\ \nu_\beta \end{bmatrix} \begin{bmatrix} \nu_\alpha & \nu_\beta \end{bmatrix} \succeq 0.$$

Both diagonal caps are active. Here the residual correlation is $r_{\alpha,\beta} = (\rho_{\mathbf{\Sigma}} - \nu_\alpha \nu_\beta)/\sqrt{v_1 v_2}$. The $\mathcal{P}_1$ inequalities give $0 \le r_{\alpha,\beta} < 1$, as verified in Appendix B; therefore $|\widetilde{\mathbf{\Sigma}}^*_{\mathrm{N}}| = v_1 v_2 (1 - r^2_{\alpha,\beta}) > 0$.

On $\mathcal{P}_2$, the candidate is positive diagonal and both caps hold. The covariance difference has diagonal entries $\nu^2_\alpha, \nu^2_\beta$ and determinant $\nu^2_\alpha \nu^2_\beta - \rho^2_{\mathbf{\Sigma}} \ge 0$, by $\rho_{\mathbf{\Sigma}} \le \nu_\alpha \nu_\beta$. These are the principal-minor conditions for that difference to be positive semidefinite. On $\mathcal{P}_3$, subtraction instead gives

$$\widetilde{\mathbf{\Sigma}} - \widetilde{\mathbf{\Sigma}}^*_{\mathrm{N}} = (1 - v_1) \begin{bmatrix} 1 \\ \rho_{\mathbf{\Sigma}} \end{bmatrix} \begin{bmatrix} 1 & \rho_{\mathbf{\Sigma}} \end{bmatrix} \succeq 0.$$

Its first cap is active, and its second diagonal is $1 - \rho^2_{\mathbf{\Sigma}} \nu^2_\alpha \le 1 - \nu^2_\beta = v_2$, because $\rho_{\mathbf{\Sigma}} > \nu_\beta/\nu_\alpha$. Its determinant is $v_1(1 - \rho^2_{\mathbf{\Sigma}}) > 0$. The same calculations with the two coordinates interchanged prove feasibility and positive definiteness on $\mathcal{P}_4$. Congruence by the positive diagonal matrix $\mathbf{t}_{\mathbf{\Sigma}}$ preserves all these covariance-order and positive-definiteness properties and converts the normalized caps v_j to $\min\{\Sigma_{jj}, D_j e^{2r_j}\}$.

Directly expanding the determinants gives

$$|\widetilde{\mathbf{\Sigma}}^*_{\mathrm{N}}| = \begin{cases} v_1 v_2 (1 - r^2_{\alpha,\beta}), & \text{if } (\alpha, \beta) \in \mathcal{P}_1, \\ v_1 v_2, & \text{if } (\alpha, \beta) \in \mathcal{P}_2, \\ v_1 (1 - \rho^2_{\mathbf{\Sigma}}), & \text{if } (\alpha, \beta) \in \mathcal{P}_3, \\ v_2 (1 - \rho^2_{\mathbf{\Sigma}}), & \text{if } (\alpha, \beta) \in \mathcal{P}_4. \end{cases} \tag{172}$$

At the $\mathcal{P}_1$–$\mathcal{P}_2$ interface, $\rho_{\mathbf{\Sigma}} = \nu_\alpha \nu_\beta$, so both off-diagonal entries are zero. At the $\mathcal{P}_1$–$\mathcal{P}_3$ interface, $\nu_\beta = \rho_{\mathbf{\Sigma}} \nu_\alpha$, so the first candidate's off-diagonal entry is $\rho_{\mathbf{\Sigma}} v_1$ and its second diagonal is $v_2 = 1 - \rho^2_{\mathbf{\Sigma}} + \rho^2_{\mathbf{\Sigma}} v_1$. The $\mathcal{P}_1$–$\mathcal{P}_4$ interface follows by interchanging the coordinates. Hence the matrices agree on all shared finite interfaces.

If $\alpha > 0$ and $\beta = 0$, use the $\mathcal{P}_3$ matrix with $v_2 = 1$; its determinant is $v_1(1 - \rho^2_{\mathbf{\Sigma}})$ and its common-rate value is α. If $\alpha = 0$ and $\beta > 0$, use the $\mathcal{P}_4$ matrix with $v_1 = 1$, giving value β. The preceding covariance-order and cap calculations remain valid on these axes. If both deficits are zero, take $\mathbf{\Sigma}^*_{\mathrm{N}} = \mathbf{\Sigma}$, giving common rate zero. These choices agree with the limits of the positive-deficit formulas and with (28).

Since $|\mathbf{\Sigma}| = \Sigma_{11}\Sigma_{22}(1 - \rho^2_{\mathbf{\Sigma}})$ and $|\mathbf{\Sigma}^*_{\mathrm{N}}| = \Sigma_{11}\Sigma_{22}|\widetilde{\mathbf{\Sigma}}^*_{\mathrm{N}}|$, determinant scaling and (172) give

$$\frac{1}{2} \log \frac{|\mathbf{\Sigma}|}{|\mathbf{\Sigma}^*_{\mathrm{N}}|} = \frac{1}{2} \log \frac{1 - \rho^2_{\mathbf{\Sigma}}}{|\widetilde{\mathbf{\Sigma}}^*_{\mathrm{N}}|} = \Upsilon_{\rho_{\mathbf{\Sigma}}}(\alpha, \beta). \tag{173}$$

For example, the $\mathcal{P}_1$ case uses $-\frac{1}{2}\log(v_1 v_2) = \alpha + \beta$; the $\mathcal{P}_3$ case uses $-\frac{1}{2}\log v_1 = \alpha$. The other two cases give their corresponding expressions in (27).

To prove optimality, let $\mathbf{\Sigma}_{\mathrm{N}}$ be any positive-definite feasible covariance. Take independent Gaussian vectors $(\mathbf{W}, \mathbf{N})$ with covariances $(\mathbf{\Sigma} - \mathbf{\Sigma}_{\mathrm{N}}, \mathbf{\Sigma}_{\mathrm{N}})$ and let $(X, Y)^{\mathsf{T}} = \mathbf{W} + \mathbf{N}$. Gaussian entropy differences give

$$I(X, Y; \mathbf{W}) = \frac{1}{2} \log \frac{|\mathbf{\Sigma}|}{|\mathbf{\Sigma}_{\mathrm{N}}|}.$$

The diagonal caps and $\mathbf{\Sigma}_{\mathrm{N}} \preceq \mathbf{\Sigma}$ imply $\Sigma_{\mathrm{N},11} \le \min\{\Sigma_{11}, D_1 e^{2r_1}\} = \Sigma_{11} e^{-2\alpha}$, and therefore $I(X;\mathbf{W}) = \frac{1}{2}\log(\Sigma_{11}/\Sigma_{\mathrm{N},11}) \ge \alpha$. The second coordinate similarly gives $I(Y;\mathbf{W}) \ge \beta$. Yu's unrestricted envelope [13, Theorem 3] now yields

$$\frac{1}{2}\log\frac{|\mathbf{\Sigma}|}{|\mathbf{\Sigma}_{\mathrm{N}}|} \ge \Upsilon_{\rho_{\mathbf{\Sigma}}}(\alpha,\beta).$$

The candidates attain this lower bound by (173), and so maximize the determinant. A singular residual has determinant zero and cannot maximize. Strict concavity of $\log|\mathbf{\Sigma}_{\mathrm{N}}|$ on the convex positive-definite feasible set makes the maximizer unique.

We next prove the covering assertion for a fixed retained cell $q \in \mathcal{J}_n$. Use the optimizer $\mathbf{\Sigma}_{\mathrm{N},q} \in \mathcal{F}_q$ and define $\mathbf{b}_q := (\mathbf{\Sigma}_q - \mathbf{\Sigma}_{\mathrm{N},q})^{1/2}$ and $I_q := \frac{1}{2}\log(|\mathbf{\Sigma}_q|/|\mathbf{\Sigma}_{\mathrm{N},q}|)$. The bound (170) makes all constants below independent of q, the nonnegative design rates, and the shell point.

Introduce the Gaussian distribution Q under which $\mathbf{U}_q \sim \mathcal{N}(\mathbf{0}_2, \mathbf{1}_2)$ and $\mathbf{N} \sim \mathcal{N}(\mathbf{0}_2, \mathbf{\Sigma}_{\mathrm{N},q})$ are independent and $\mathbf{X} = \mathbf{b}_q\mathbf{U}_q + \mathbf{N}$. Dividing the Gaussian conditional density of $\mathbf{X}$ given $\mathbf{U}_q = \mathbf{u}$ by its marginal density gives

$$\imath(\mathbf{u};\mathbf{x}) = I_q - \frac{1}{2}(\mathbf{x}-\mathbf{b}_q\mathbf{u})^{\mathsf{T}}\mathbf{\Sigma}_{\mathrm{N},q}^{-1}(\mathbf{x}-\mathbf{b}_q\mathbf{u}) + \frac{1}{2}\mathbf{x}^{\mathsf{T}}\mathbf{\Sigma}_q^{-1}\mathbf{x}.$$

Here the determinant ratio in I_q is the ratio of the two Gaussian normalizing factors. Completing the square in the joint density, or applying Gaussian conditioning, gives

$$\mathbf{U}_q \mid \mathbf{X} = \mathbf{x} \sim \mathcal{N}(\mathbf{b}_q^{\mathsf{T}}\mathbf{\Sigma}_q^{-1}\mathbf{x}, \mathbf{\Sigma}_{\mathbf{U}|\mathbf{X},q}),$$

where

$$\mathbf{\Sigma}_{\mathbf{U}|\mathbf{X},q} = \mathbf{1}_2 - \mathbf{b}_q^{\mathsf{T}}\mathbf{\Sigma}_q^{-1}\mathbf{b}_q = (\mathbf{1}_2 + \mathbf{b}_q^{\mathsf{T}}\mathbf{\Sigma}_{\mathrm{N},q}^{-1}\mathbf{b}_q)^{-1} \succ 0.$$

The last equality is the matrix inversion identity with $\mathbf{\Sigma}_q = \mathbf{\Sigma}_{\mathrm{N},q} + \mathbf{b}_q\mathbf{b}_q^{\mathsf{T}}$. Its inverse expression proves positive definiteness even if $\mathbf{b}_q$ is singular.

For a fixed shell point $\mathbf{x}^n$, use the product posterior $Q(\cdot \mid \mathbf{x}^n)$. Define $J := \sum_{i\in[n]} \imath(\mathbf{U}_{q,i};\mathbf{x}_i)$ and $E_j := \Delta_j(\mathbf{U}_q^n;\mathbf{x}^n)$, where $\Delta_j(\mathbf{u}^n;\mathbf{x}^n) := \|(\mathbf{x}^n - \mathbf{b}_q\mathbf{u}^n)_j\|^2$. The following lemma concerns this posterior distribution; it does not depend on a selected codebook.

Lemma 9 (Uniform posterior small-ball bound)**.** *Let $\boldsymbol{Y} := (J, E_1, E_2)^{\mathsf{T}}$ under the product posterior at a fixed source block. There exist $c > 0$ and an integer n_0 such that, for every $n \ge n_0$,*

$$Q\left\{\|\boldsymbol{Y} - \mathsf{E}_Q[\boldsymbol{Y} \mid \mathbf{x}^n]\| \le \frac{s_n}{64} \,\middle|\, \mathbf{x}^n\right\} \ge \frac{c}{(\log n)^7}, \tag{174}$$

uniformly over $q \in \mathcal{J}_n$, all nonnegative design rates, and $\mathbf{x}^n \in \mathcal{T}_n(\mathbf{\Sigma}_q)$.

Its proof follows the covering argument below.

Write $\widehat{\mathbf{\Sigma}} := \widehat{\mathbf{\Sigma}}(\mathbf{x}^n)$ and let $\boldsymbol{G}$ be the Gram matrix of the posterior residual block $\mathbf{x}^n - \mathbf{b}_q\mathbf{U}_q^n$. For each column, Gaussian conditioning gives residual mean $\mathbf{\Sigma}_{\mathrm{N},q}\mathbf{\Sigma}_q^{-1}\mathbf{x}$ and covariance $\mathbf{\Sigma}_{\mathrm{N},q} - \mathbf{\Sigma}_{\mathrm{N},q}\mathbf{\Sigma}_q^{-1}\mathbf{\Sigma}_{\mathrm{N},q}$. Since $\mathbf{x}^n(\mathbf{x}^n)^{\mathsf{T}} = n\widehat{\mathbf{\Sigma}}$,

$$\boldsymbol{G}_n := \mathsf{E}_Q[\boldsymbol{G} \mid \mathbf{x}^n] = n\mathbf{\Sigma}_{\mathrm{N},q} + n\mathbf{\Sigma}_{\mathrm{N},q}\mathbf{\Sigma}_q^{-1}(\widehat{\mathbf{\Sigma}} - \mathbf{\Sigma}_q)\mathbf{\Sigma}_q^{-1}\mathbf{\Sigma}_{\mathrm{N},q}. \tag{175}$$

Its j-th diagonal entry is $\mathsf{E}_Q[E_j \mid \mathbf{x}^n]$. Summing the information densities gives

$$J = nI_q - \frac{1}{2}\operatorname{tr}(\mathbf{\Sigma}_{\mathrm{N},q}^{-1}\boldsymbol{G}) + \frac{n}{2}\operatorname{tr}(\mathbf{\Sigma}_q^{-1}\widehat{\mathbf{\Sigma}}), \tag{176}$$

and hence

$$\mathsf{E}_Q[J \mid \mathbf{x}^n] = nI_q + \frac{n}{2}\operatorname{tr}[\mathbf{\Sigma}_q^{-1}(\mathbf{\Sigma}_q - \mathbf{\Sigma}_{\mathrm{N},q})\mathbf{\Sigma}_q^{-1}(\widehat{\mathbf{\Sigma}} - \mathbf{\Sigma}_q)].$$

The uniform operator bounds (170), trace Cauchy–Schwarz, and the shell condition $\left\|\widehat{\mathbf{\Sigma}} - \mathbf{\Sigma}_q\right\| \le C_0/n$ imply

$$|\mathsf{E}_Q[J \mid \mathbf{x}^n] - nI_q| + \sum_{j\in[2]} |\mathsf{E}_Q[E_j \mid \mathbf{x}^n] - n[\mathbf{\Sigma}_{\mathrm{N},q}]_{jj}| \le C,$$

with C independent of the cell, rates, and shell point.

Define the simultaneous posterior event

$$\mathcal{G}_n(\mathbf{x}^n) := \{\mathbf{u}^n : \imath(\mathbf{u}^n;\mathbf{x}^n) \le nI_q + s_n/16, \quad \Delta_j(\mathbf{u}^n;\mathbf{x}^n) \le n[\mathbf{\Sigma}_{\mathrm{N},q}]_{jj} + s_n/16 \text{ for } j \in [2]\}.$$

On the small-ball event in Lemma 9, each coordinate differs from its posterior mean by at most $s_n/64$. Since $s_n \to \infty$, the preceding constant is at most $3s_n/64$ for sufficiently large n. Thus that event is contained in $\mathcal{G}_n(\mathbf{x}^n)$, and its posterior probability is at least $c(\log n)^{-7}$.

Bayes' identity gives $dQ_{\mathbf{U}_q}^{\otimes n}/dQ(\cdot \mid \mathbf{x}^n) = e^{-J}$. This identity is valid for every fixed $\mathbf{x}^n$, because both Gaussian densities are positive. Hence

$$\begin{aligned} Q_{\mathbf{U}_q}^{\otimes n}\{\mathcal{G}_n(\mathbf{x}^n)\} &= \mathsf{E}_Q[e^{-J}\mathbb{1}_{\mathcal{G}_n(\mathbf{x}^n)} \mid \mathbf{x}^n] \\ &\geq e^{-nI_q - s_n/16} Q\{\mathcal{G}_n(\mathbf{x}^n) \mid \mathbf{x}^n\} \\ &\geq c(\log n)^{-7} e^{-nI_q - s_n/16} =: p_n. \end{aligned}$$

Generate $M_{0,q} = \lceil e^{nI_q + s_n/8} \rceil$ independent standard Gaussian codewords. For a fixed shell point, independence and $1 - t \leq e^{-t}$ give the missed-covering bound

$$(1 - p_n)^{M_{0,q}} \leq e^{-p_n M_{0,q}} \leq \exp\{-c(\log n)^{-7} e^{s_n/16}\}.$$

The last inequality uses $M_{0,q} \geq e^{nI_q + s_n/8}$, which leaves the positive exponent slack $s_n/16$ after the change of measure.

To cover the entire shell, take a maximal δ_n-separated set[1] $\mathcal{N}_n \subset \mathcal{T}_n(\mathbf{\Sigma}_q)$, where $\delta_n := 1/\log n$. The shell is closed and bounded, because

$$\|\mathbf{x}^n\|^2 = n \operatorname{tr} \widehat{\mathbf{\Sigma}} \leq n \operatorname{tr} \mathbf{\Sigma}_q + \sqrt{2} C_0 \leq C^2 n.$$

The first inequality uses $|\operatorname{tr}(\widehat{\mathbf{\Sigma}} - \mathbf{\Sigma}_q)| \leq \sqrt{2} \left\| \widehat{\mathbf{\Sigma}} - \mathbf{\Sigma}_q \right\|$. The shell is therefore compact and admits a finite maximal separated set. Maximality makes it a δ_n-net[2]: otherwise, a point farther than δ_n from all selected points could be added. Disjoint open balls of radius $\delta_n/2$ around its points lie inside the radius-$C\sqrt{n} + \delta_n/2$ ball in $\mathbb{R}^{2n}$. Comparing volumes gives

$$|\mathcal{N}_n| \leq \left(\frac{C\sqrt{n} + \delta_n/2}{\delta_n/2} \right)^{2n} = (1 + 2C\sqrt{n} \log n)^{2n} \leq e^{C' n \log n}.$$

The union bound over this deterministic net consequently bounds the probability of missing at least one net point by

$$\exp\{C' n \log n - c(\log n)^{-7} e^{s_n/16}\} \longrightarrow 0.$$

Indeed, the logarithm of the ratio of the second term in the exponent to $n \log n$ is $s_n/16 - \log n - 8 \log\log n + O(1)$, which tends to infinity because $\sqrt{n}/(\log n)^2 \to \infty$. For sufficiently large n, a codebook realization therefore exists that covers every net point simultaneously.

Fix such a realization. Given $\mathbf{x}^n \in \mathcal{T}_n(\mathbf{\Sigma}_q)$, choose $\bar{\mathbf{x}}^n \in \mathcal{N}_n$ with $\|\mathbf{x}^n - \bar{\mathbf{x}}^n\| \leq \delta_n$ and a common codeword good for $\bar{\mathbf{x}}^n$. The rowwise triangle inequality gives

$$\begin{aligned} \|(\mathbf{x}^n - \mathbf{b}_q \mathbf{u}^n)_j\|^2 &\leq \left(\sqrt{n[\mathbf{\Sigma}_{\mathrm{N},q}]_{jj} + s_n/16} + \delta_n \right)^2 \\ &= n[\mathbf{\Sigma}_{\mathrm{N},q}]_{jj} + s_n/16 + 2\delta_n \sqrt{n[\mathbf{\Sigma}_{\mathrm{N},q}]_{jj} + s_n/16} + \delta_n^2 \\ &\leq n[\mathbf{\Sigma}_{\mathrm{N},q}]_{jj} + (17/16 + 2\sqrt{\kappa^{-1} + 1}) s_n \\ &\leq n[\mathbf{\Sigma}_{\mathrm{N},q}]_{jj} + \chi_\kappa s_n = \xi_{j,q,n}^2. \end{aligned}$$

The penultimate inequality uses $[\mathbf{\Sigma}_{\mathrm{N},q}]_{jj} \leq \kappa^{-1}$, $s_n/n \leq 1$, $\delta_n\sqrt{n} = s_n$, and $\delta_n^2 \leq s_n$; the last uses (69). Thus every shell point, rather than only every net point, has a common codeword leaving both residuals inside the prescribed private balls.

Apply Lemma 5 to these balls. It gives deterministic private offset codebooks satisfying

$$\begin{aligned} \log M_{j,q} &\leq \log C_{\mathrm{cov}} + 4 \log n + \frac{n}{2} \left[\log \frac{[\mathbf{\Sigma}_{\mathrm{N},q}]_{jj} + \chi_\kappa s_n/n}{D_j} \right]^+ \\ &\leq \frac{n}{2} \left[\log \frac{[\mathbf{\Sigma}_{\mathrm{N},q}]_{jj}}{D_j} \right]^+ + \frac{\chi_\kappa}{2\kappa} s_n + 4 \log n + \log C_{\mathrm{cov}}. \end{aligned}$$

For the second inequality, write the logarithm as $\log([\mathbf{\Sigma}_{\mathrm{N},q}]_{jj}/D_j) + \log(1 + \chi_\kappa s_n/(n[\mathbf{\Sigma}_{\mathrm{N},q}]_{jj}))$, use $[a+b]^+ \leq [a]^+ + b$ for $b \geq 0$, and then use $\log(1+t) \leq t$ and $[\mathbf{\Sigma}_{\mathrm{N},q}]_{jj} \geq \kappa$. Each actual residual is covered by one of these offsets, so the decoded block has squared error at most nD_j and hence block distortion at most D_j. The local maps f_q and $\phi_{j,q}$ in Section IV-B choose such a common codeword and private offset, so this is the distortion guarantee in (83). Finally, $I_q \geq 0$ because $\mathbf{\Sigma}_{\mathrm{N},q} \preceq \mathbf{\Sigma}_q$, and $\lceil e^t \rceil \leq 2e^t$ for $t \geq 0$ gives $\log M_{0,q} \leq nI_q + s_n/8 + \log 2$. Enlarging the common constant proves (84)–(85). The common gain $\mathbf{b}_q$ was never inverted, so the argument includes rank-deficient $\mathbf{\Sigma}_q - \mathbf{\Sigma}_{\mathrm{N},q}$.

[1] A δ-separated set has pairwise distances at least δ.

[2] A δ-net of a set is a collection of centers whose radius-δ balls cover that set.

A. Proof of Lemma 9

Recall the Gaussian posterior and its positive-definite covariance $\mathbf{\Sigma}_{\mathbf{U}|\mathbf{X},q}$ from the proof above. Let $\mathbf{Z}^n$ have independent standard Gaussian entries. The product posterior can be represented as

$$\mathbf{U}_q^n = \mathbf{b}_q^\mathsf{T}\mathbf{\Sigma}_q^{-1}\mathbf{x}^n + \mathbf{\Sigma}_{\mathbf{U}|\mathbf{X},q}^{1/2}\mathbf{Z}^n.$$

The fixed shell point satisfies $\widehat{\mathbf{\Sigma}} \succeq (\kappa - C_0/n)\mathbf{1}_2 \succeq (\kappa/2)\mathbf{1}_2$ for sufficiently large n. Thus $n^{-1/2}\widehat{\mathbf{\Sigma}}^{-1/2}\mathbf{x}^n$ has two orthonormal rows. Extend those rows to an orthogonal basis of $\mathbb{R}^n$ to choose an orthogonal matrix O such that

$$\mathbf{x}^n O = \sqrt{n}\,\widehat{\mathbf{\Sigma}}^{1/2}[\mathbf{1}_2\ \mathbf{0}_{2\times(n-2)}].$$

Right multiplication by O preserves every residual Gram matrix, and $\mathbf{Z}^n O$ has the same distribution as $\mathbf{Z}^n$ by orthogonal invariance of independent standard Gaussian rows. We may therefore use the displayed form for $\mathbf{x}^n$ without changing the conditional distribution of $\boldsymbol{G}$, J, or (E_1, E_2).

Write $\mathbf{Z}^n = [A\ B]$, where $A \in \mathbb{R}^{2\times 2}$ and $B \in \mathbb{R}^{2\times(n-2)}$ are independent standard Gaussian matrices. Define the deterministic matrices $H := \mathbf{\Sigma}_{\mathrm{N},q}\mathbf{\Sigma}_q^{-1}$ and $D := \mathbf{b}_q\mathbf{\Sigma}_{\mathbf{U}|\mathbf{X},q}^{1/2}$. Here D is a gain matrix, whereas D_j denotes a scalar distortion level. The posterior residual is

$$\mathbf{N}^n := \mathbf{x}^n - \mathbf{b}_q\mathbf{U}_q^n = H\mathbf{x}^n - D\mathbf{Z}^n.$$

This equality follows from $\mathbf{1}_2 - \mathbf{b}_q\mathbf{b}_q^\mathsf{T}\mathbf{\Sigma}_q^{-1} = \mathbf{\Sigma}_{\mathrm{N},q}\mathbf{\Sigma}_q^{-1}$. All coefficient norms are uniformly bounded: for example, $\|H\|_{\mathrm{op}} \le \kappa^{-2}$, $\|D\|_{\mathrm{op}} \le \kappa^{-1/2}$, and $\left\|\mathbf{\Sigma}_{\mathrm{N},q}^{-1}\right\| \le \sqrt{2}\kappa^{-1}$. The bound for D uses $\mathbf{\Sigma}_{\mathbf{U}|\mathbf{X},q} \preceq \mathbf{1}_2$ and $\mathbf{b}_q\mathbf{b}_q^\mathsf{T} \preceq \mathbf{\Sigma}_q \preceq \kappa^{-1}\mathbf{1}_2$.

Expanding $\boldsymbol{G} = \mathbf{N}^n(\mathbf{N}^n)^\mathsf{T}$ and subtracting its expectation gives

$$\begin{aligned}\boldsymbol{G} - \boldsymbol{G}_n = &-H\mathbf{x}^n(\mathbf{Z}^n)^\mathsf{T}D^\mathsf{T} - D\mathbf{Z}^n(\mathbf{x}^n)^\mathsf{T}H^\mathsf{T} \\ &+ D[\mathbf{Z}^n(\mathbf{Z}^n)^\mathsf{T} - n\mathbf{1}_2]D^\mathsf{T}. \end{aligned} \tag{177}$$

The two cross terms have expectation zero, since $\mathsf{E}[\mathbf{Z}^n] = 0$, and the final centering uses $\mathsf{E}[\mathbf{Z}^n(\mathbf{Z}^n)^\mathsf{T}] = n\mathbf{1}_2$. In addition, the diagonal entries of $\boldsymbol{G} - \boldsymbol{G}_n$ are the centered residual errors, while (176) gives $J - \mathsf{E}_Q[J \mid \mathbf{x}^n] = -\frac{1}{2}\operatorname{tr}[\mathbf{\Sigma}_{\mathrm{N},q}^{-1}(\boldsymbol{G} - \boldsymbol{G}_n)]$. Hence the trace Cauchy–Schwarz inequality implies

$$\|\boldsymbol{Y} - \mathsf{E}_Q[\boldsymbol{Y} \mid \mathbf{x}^n]\| \le \sqrt{1 + \tfrac{1}{4}\left\|\mathbf{\Sigma}_{\mathrm{N},q}^{-1}\right\|^2}\ \|\boldsymbol{G} - \boldsymbol{G}_n\|.$$

This bound does not require the covariance of $\boldsymbol{Y}$ to be nonsingular.

For a fixed $a \in (0,1)$, to be chosen using the uniform coefficient bounds, define

$$\mathcal{E}_n := \{\|A\| \le a/\log n, \quad \left\|BB^\mathsf{T} - (n-2)\mathbf{1}_2\right\| \le as_n\}.$$

On this event, the rotated form of $\mathbf{x}^n$ gives

$$\left\|\mathbf{x}^n(\mathbf{Z}^n)^\mathsf{T}\right\| = \sqrt{n}\left\|\widehat{\mathbf{\Sigma}}^{1/2}A^\mathsf{T}\right\| \le C\sqrt{n}\,\|A\| \le Cas_n.$$

Also, splitting $\mathbf{Z}^n$ into its two blocks gives

$$\begin{aligned}\left\|\mathbf{Z}^n(\mathbf{Z}^n)^\mathsf{T} - n\mathbf{1}_2\right\| &\le \left\|AA^\mathsf{T} - 2\mathbf{1}_2\right\| + \left\|BB^\mathsf{T} - (n-2)\mathbf{1}_2\right\| \\ &\le \|A\|^2 + 2\sqrt{2} + as_n \le C + as_n.\end{aligned}$$

The inequality $\left\|AA^\mathsf{T}\right\| \le \|A\|^2$ follows from the Frobenius/operator norm product bound. Combining these estimates with (177) and the preceding bound for $\boldsymbol{Y}$ yields

$$\|\boldsymbol{Y} - \mathsf{E}_Q[\boldsymbol{Y} \mid \mathbf{x}^n]\| \le C_1as_n + C_2,$$

with constants independent of the retained cell and shell point. Choose $a > 0$ so small that $C_1a \le 1/128$, and then choose n large enough that $C_2 \le s_n/128$. This proves that $\mathcal{E}_n$ is contained in the small-ball event in (174).

It remains to bound the probability of $\mathcal{E}_n$ from below. Regard A as a four-dimensional standard Gaussian vector. Its density is at least $(2\pi)^{-2}e^{-1/2}$ on the unit ball. Integrating this lower bound over the radius-$a/\log n$ ball gives

$$\Pr\{\|A\| \le a/\log n\} \ge c_a(\log n)^{-4},$$

because four-dimensional ball volume scales as the fourth power of its radius.

Put $m := n - 2$ and define the independent identically distributed vectors

$$V_i := (B_{1i}^2 - 1, B_{1i}B_{2i}, B_{2i}^2 - 1)^\mathsf{T}, \qquad i \in [m].$$

They have mean zero and covariance $\mathbf{\Sigma}_V := \operatorname{diag}(2,1,2)$. The diagonal variances follow from $\mathsf{E}[Z^4] = 3$ and independence of B_{1i}, B_{2i}; every off-diagonal covariance vanishes by independence or an odd Gaussian moment. The third absolute moment of

$\mathbf{\Sigma}_V^{-1/2} V_i$ is finite because its entries are quadratic polynomials of Gaussian variables. Therefore the convex-set Berry–Esseen bound [16, Theorem 1.1], applied after this fixed whitening, gives a constant C such that

$$\left| \Pr\left\{ m^{-1/2} \sum_{i\in[m]} V_i \in \mathcal{A} \right\} - \Pr\{\mathbf{Z}_{\mathrm{G}} \in \mathcal{A}\} \right| \le C/\sqrt{m}$$

for every convex Borel set $\mathcal{A} \subset \mathbb{R}^3$, where $\mathbf{Z}_{\mathrm{G}} \sim \mathcal{N}(\mathbf{0}_3, \mathbf{\Sigma}_V)$. The bound is uniform in the set, so it applies to a shrinking ellipsoid.

Specifically, the three independent entries of $BB^{\mathsf{T}} - m\mathbf{1}_2$ are the coordinates of $\sum_i V_i$. Its squared Frobenius norm counts the off-diagonal entry twice. Thus its radius-as_n event is exactly the preceding normalized-sum event with

$$\mathcal{A}_{n,a} := \{v \in \mathbb{R}^3 : v_1^2 + 2v_2^2 + v_3^2 \le a^2 s_n^2/m\}.$$

Let $t_n := as_n/\sqrt{m}$. For sufficiently large n, $t_n \le 1$, so this ellipsoid lies in the unit ball, on which the Gaussian density of $\mathbf{Z}_{\mathrm{G}}$ has a positive minimum. Its volume is $(4\pi/(3\sqrt{2}))t_n^3$, by the linear change of variables $(v_1, v_2, v_3) \mapsto (v_1, \sqrt{2}v_2, v_3)$. Since $s_n/\sqrt{m} = \sqrt{n/(n-2)}/\log n \ge 1/\log n$, this gives

$$\Pr\{\mathbf{Z}_{\mathrm{G}} \in \mathcal{A}_{n,a}\} \ge c'_a (\log n)^{-3}.$$

Consequently,

$$\begin{aligned}\Pr\{\left\|BB^{\mathsf{T}} - m\mathbf{1}_2\right\| \le as_n\} &\ge c'_a(\log n)^{-3} - C/\sqrt{m} \\ &\ge \tfrac{1}{2} c'_a (\log n)^{-3}\end{aligned}$$

for sufficiently large n, because $(\log n)^3/\sqrt{n-2} \to 0$. Finally, A and B consist of disjoint independent Gaussian columns. Multiplying the two probability lower bounds therefore gives $Q(\mathcal{E}_n \mid \mathbf{x}^n) \ge c(\log n)^{-7}$. Together with the event inclusion already proved, this establishes (174). The Gaussian distributions in these probability estimates are fixed; the retained cell and shell point enter only through uniformly bounded deterministic coefficients. All constants and the threshold n_0 can consequently be chosen uniformly, including when the common covariance loses rank.

Appendix F
Proof of (116)

Fix $\zeta > 0$. For a positive-definite $\mathbf{\Sigma}$ and admissible private rates, define the displacement and Taylor remainder

$$\begin{aligned}\Delta(\mathbf{\Sigma}, r_1, r_2) &:= \|\mathbf{\Sigma} - \mathbf{\Sigma}_{XY}\| + \sum_{j\in[2]} |r_j - R_j^*|, \\ \psi(\mathbf{\Sigma}, r_1, r_2) &:= \mathsf{R}_0(r_1, r_2, D_1, D_2 \mid Q_{XY}^{\mathbf{\Sigma}}) - R_0^* - \langle \mathbf{g}, \mathbf{\Sigma} - \mathbf{\Sigma}_{XY}\rangle + \sum_{j\in[2]} \lambda_j^*(r_j - R_j^*).\end{aligned}$$

Proposition 1 and (105) imply that, for every $a > 0$, some $\delta_a > 0$ satisfies

$$|\psi(\mathbf{\Sigma}, r_1, r_2)| \le a\Delta(\mathbf{\Sigma}, r_1, r_2) \tag{178}$$

whenever $\Delta \le \delta_a$. At a zero first-order private rate this uses the one-sided derivative, which suffices because the design rates are nonnegative.

Fix $B > 0$, and define $\mathcal{G}_{n,B} := \{\sqrt{n} \left\|\hat{\mathbf{\Sigma}}_n - \mathbf{\Sigma}_{XY}\right\| \le B\}$ and $C_B := B + |L_1| + |L_2|$. For all sufficiently large n, $\mathcal{G}_{n,B}$ implies $\hat{\mathbf{\Sigma}}_n \in \mathcal{K}_0$, because $B/\sqrt{n} \le \eta_{\mathrm{cov}}$. Moreover, the private-rate choices (95) give exactly

$$\sqrt{n}(r_{j,n} - R_j^*) = L_j, \qquad j \in [2],$$

and both rates are nonnegative eventually: this is immediate when $R_j^* > 0$, while $R_j^* = 0$ requires $L_j \ge 0$ by Theorem 1. Consequently, on $\mathcal{G}_{n,B}$,

$$\Delta(\hat{\mathbf{\Sigma}}_n, r_{1,n}, r_{2,n}) \le \frac{C_B}{\sqrt{n}}, \tag{179}$$

$$E_n = \sqrt{n}\psi(\hat{\mathbf{\Sigma}}_n, r_{1,n}, r_{2,n}). \tag{180}$$

The second equality follows by substituting (111) and the displayed private-rate offsets: their two linear terms cancel exactly. Choose $a := \zeta/(2C_B)$. For all sufficiently large n, $C_B/\sqrt{n} \le \delta_a$; hence (178) gives

$$|E_n| \le \sqrt{n}\, a\Delta(\hat{\mathbf{\Sigma}}_n, r_{1,n}, r_{2,n}) \le aC_B = \zeta/2 < \zeta$$

on $\mathcal{G}_{n,B}$. Thus $\{|E_n| > \zeta\} \subseteq \mathcal{G}_{n,B}^{\mathrm{c}}$ for all sufficiently large n. We may define $E_n = 0$ outside $\mathcal{K}_0$, since (113) already includes that event in a separate probability term.

To control the complement, put $\mathbf{H}_i := \mathbf{X}_i\mathbf{X}_i^{\mathsf{T}} - \mathbf{\Sigma}_{XY}$. The matrices $\mathbf{H}_i$ are independent and centered. The Gaussian fourth-moment identity $\mathsf{E}[X_a^2X_b^2] = [\mathbf{\Sigma}_{XY}]_{aa}[\mathbf{\Sigma}_{XY}]_{bb} + 2[\mathbf{\Sigma}_{XY}]_{ab}^2$ gives

$$\begin{aligned}\mathsf{E}_{P_{XY}^n}\big[n\left\|\hat{\mathbf{\Sigma}}_n - \mathbf{\Sigma}_{XY}\right\|^2\big] &= \frac{1}{n}\sum_{i\in[n]}\mathsf{E}_{P_{XY}}\|\mathbf{H}_i\|^2 \\ &= \mathsf{E}_{P_{XY}}[(\mathbf{X}^{\mathsf{T}}\mathbf{X})^2] - \operatorname{tr}(\mathbf{\Sigma}_{XY}^2) \\ &= (\operatorname{tr}\mathbf{\Sigma}_{XY})^2 + \operatorname{tr}(\mathbf{\Sigma}_{XY}^2) =: C. \end{aligned} \tag{181}$$

Markov's inequality therefore yields

$$P_{XY}^n(\mathcal{G}_{n,B}^{\mathrm{c}}) = P_{XY}^n\{n\left\|\hat{\mathbf{\Sigma}}_n - \mathbf{\Sigma}_{XY}\right\|^2 > B^2\} \le \frac{C}{B^2}. \tag{182}$$

Since B and ζ are fixed while $n\to\infty$, the preceding event inclusion implies

$$0 \le \limsup_{n\to\infty} P_{XY}^n\{|E_n| > \zeta\} \le \frac{C}{B^2}. \tag{183}$$

This holds for every $B>0$, and the left-hand side does not depend on B. Letting $B\to\infty$ *after* taking the limit superior in n proves (116).

Appendix G
Proof of Lemma 8

Fix $\boldsymbol{\lambda}^* = (\lambda_1^*, \lambda_2^*) \in (0,1]^2$. By Yu's unrestricted Gaussian information envelope [13, Theorem 3],

$$\Upsilon_{\rho_{XY}}(\alpha,\beta) = \inf_{Q_{W|XY}:\ I_Q(X;W)\ge\alpha,\ I_Q(Y;W)\ge\beta} I_Q(X,Y;W), \qquad \alpha,\beta\ge 0.$$

This envelope is convex: independently time sharing two auxiliary kernels, with the time-sharing variable included in W, convexly combines $I(X;W)$, $I(Y;W)$, and $I(X,Y;W)$. The tangent plane at the strictly positive (α^*,β^*) is therefore a global lower bound with slopes $(\lambda_1^*,\lambda_2^*)$ by Proposition 1.

For $\mathbf{0}_{2\times 2} \prec \mathbf{\Sigma}_{\mathrm{N}} \preceq \mathbf{\Sigma}_{XY}$, set

$$\begin{aligned}\alpha_{\mathbf{\Sigma}_{\mathrm{N}}} &:= \frac{1}{2}\log\frac{\sigma_X^2}{\Sigma_{\mathrm{N},11}}, \\ \beta_{\mathbf{\Sigma}_{\mathrm{N}}} &:= \frac{1}{2}\log\frac{\sigma_Y^2}{\Sigma_{\mathrm{N},22}}, \\ t_{\mathbf{\Sigma}_{\mathrm{N}}} &:= \frac{1}{2}\log\frac{|\mathbf{\Sigma}_{XY}|}{|\mathbf{\Sigma}_{\mathrm{N}}|}.\end{aligned}$$

The independent Gaussian decomposition $(X,Y)^{\mathsf{T}} = \mathbf{W}+\mathbf{N}$ with residual covariance $\mathbf{\Sigma}_{\mathrm{N}}$ realizes these three mutual informations, even if the common covariance is singular. Hence

$$\begin{aligned} t_{\mathbf{\Sigma}_{\mathrm{N}}} &\ge \Upsilon_{\rho_{XY}}(\alpha_{\mathbf{\Sigma}_{\mathrm{N}}}, \beta_{\mathbf{\Sigma}_{\mathrm{N}}}) \\ &\ge R_0^* + \lambda_1^*(\alpha_{\mathbf{\Sigma}_{\mathrm{N}}} - \alpha^*) + \lambda_2^*(\beta_{\mathbf{\Sigma}_{\mathrm{N}}} - \beta^*). \end{aligned} \tag{184}$$

Rearranging (184), using (32), gives for every such residual covariance

$$\begin{aligned}\frac{1}{2}\log\frac{|\mathbf{\Sigma}_{XY}|}{|\mathbf{\Sigma}_{\mathrm{N}}|} + \sum_{j\in[2]}\frac{\lambda_j^*}{2}\log\frac{\Sigma_{\mathrm{N},jj}}{D_j} &= t_{\mathbf{\Sigma}_{\mathrm{N}}} - \lambda_1^*\alpha_{\mathbf{\Sigma}_{\mathrm{N}}} - \lambda_2^*\beta_{\mathbf{\Sigma}_{\mathrm{N}}} + \sum_{j\in[2]}\frac{\lambda_j^*}{2}\log\frac{[\mathbf{\Sigma}_{XY}]_{jj}}{D_j} \\ &\ge R_0^* + \lambda_1^*R_1^* + \lambda_2^*R_2^*. \end{aligned} \tag{185}$$

At the optimizer $\mathbf{\Sigma}_{\mathrm{N}}^*$ from Appendix E, the private caps imply $\alpha_{\mathbf{\Sigma}_{\mathrm{N}}^*} \ge \alpha^*$ and $\beta_{\mathbf{\Sigma}_{\mathrm{N}}^*} \ge \beta^*$, while (173) gives $t_{\mathbf{\Sigma}_{\mathrm{N}}^*} = R_0^*$. Since both slopes are positive, (184) forces both deficits to equal their selected values. Hence equality is attained in (185), proving (128), including when a private first-order rate is zero.

For nearby $\mathbf{\Sigma}$, evaluate the infimum (124) at its common-rate residual optimizer $\mathbf{\Sigma}_{\mathrm{N},\mathbf{\Sigma}}^*$. The diagonal caps give

$$\begin{aligned}\underline{F}_{\boldsymbol{\lambda}^*}(\mathbf{\Sigma}) &\le \frac{1}{2}\log\frac{|\mathbf{\Sigma}|}{|\mathbf{\Sigma}_{\mathrm{N},\mathbf{\Sigma}}^*|} + \sum_{j\in[2]}\frac{\lambda_j^*}{2}\log\frac{[\mathbf{\Sigma}_{\mathrm{N},\mathbf{\Sigma}}^*]_{jj}}{D_j} \\ &\le \mathsf{R}_0(R_1^*, R_2^*, D_1, D_2 \mid Q_{XY}^{\mathbf{\Sigma}}) + \lambda_1^*R_1^* + \lambda_2^*R_2^*. \end{aligned} \tag{186}$$

Equality holds at $\mathbf{\Sigma}_{XY}$ by (128), but need not hold nearby. The entropy-support objective is finite on $\mathbb{S}^2_{++}$. Indeed, Hadamard's determinant bound, $\Sigma_{\mathrm{N},jj} \le \Sigma_{jj}$, and $0 < \lambda^*_j \le 1$ give

$$\underline{F}_{\boldsymbol{\lambda}^*}(\mathbf{\Sigma}) \ge \frac{1}{2}\log|\mathbf{\Sigma}| - \sum_{j\in[2]} \frac{1-\lambda^*_j}{2}\log\Sigma_{jj} - \sum_{j\in[2]} \frac{\lambda^*_j}{2}\log D_j > -\infty,$$

while taking $\mathbf{\Sigma}_{\mathrm{N}} = \mathbf{\Sigma}$ gives a finite upper bound.

By [12, Proposition 5], the function $\varphi_{\boldsymbol{\lambda}^*,\mathbf{\Sigma}_{XY}}$ in (125) is concave. It is finite on the open positive-definite cone and therefore has a supergradient $\mathbf{g}_{\mathrm{ent},\boldsymbol{\lambda}^*}$ at $\mathbf{\Sigma}_{XY}$. Negating (186) and adding the reference terms of (125) gives a differentiable lower touching function with gradient $-\mathbf{g}$: the derivatives of $\frac{1}{2}\log|\mathbf{\Sigma}|$ and $-\frac{1}{2}\operatorname{tr}(\mathbf{\Sigma}^{-1}_{XY}\mathbf{\Sigma})$ cancel at $\mathbf{\Sigma}_{XY}$. Together with (126), for every symmetric direction $\mathbf{h}$ and $t \downarrow 0$ this yields

$$-\langle \mathbf{g}, \mathbf{h}\rangle + o(1) \le \frac{\varphi_{\boldsymbol{\lambda}^*,\mathbf{\Sigma}_{XY}}(\mathbf{\Sigma}_{XY}+t\mathbf{h}) - \varphi_{\boldsymbol{\lambda}^*,\mathbf{\Sigma}_{XY}}(\mathbf{\Sigma}_{XY})}{t} \le \langle \mathbf{g}_{\mathrm{ent},\boldsymbol{\lambda}^*}, \mathbf{h}\rangle .$$

Applying the same inequality to $-\mathbf{h}$ gives $\mathbf{g}_{\mathrm{ent},\boldsymbol{\lambda}^*} = -\mathbf{g}$. Because this holds for an arbitrary supergradient, it is unique. The lower touching function and this supergradient also sandwich every sufficiently small covariance increment $\mathbf{h}$ between $-\langle \mathbf{g}, \mathbf{h}\rangle + o(\|\mathbf{h}\|)$ and $-\langle \mathbf{g}, \mathbf{h}\rangle$, proving Fréchet differentiability of φ and hence (129).

Finally, substituting $\mathbf{g}_{\mathrm{ent},\boldsymbol{\lambda}^*} = -\mathbf{g}$ in (127) gives

$$g_{\boldsymbol{\lambda}^*}(\mathbf{x}) = -\mathbf{x}^{\mathsf{T}}\mathbf{g}\mathbf{x} + \operatorname{tr}(\mathbf{g}\mathbf{\Sigma}_{XY}) = -[\jmath(x,y) - R^*_0].$$

The fluctuation is centered by Lemma 4 and has variance V by (43), completing the proof.

Appendix H
Proof of Corollary 2

On $\mathcal{P}_1$, recall $r = r_{\alpha^*,\beta^*}$ from (22), evaluated at $\rho = \rho_{XY}$, and u from (44). Evaluating the derivatives (152)–(153) in (38) gives

$$\lambda^*_1 = \frac{1-ru}{1-r^2}. \tag{187}$$

$$\lambda^*_2 = \frac{1-r/u}{1-r^2}. \tag{188}$$

On $\mathcal{P}_1$, the slope formulas (187)–(188) and the dispersion identity (47) imply (49) by substitution into (48). On $(\mathcal{P}_2, \mathcal{P}_3, \mathcal{P}_4)$, the derivative values in (155) are $(1,1)$, $(1,0)$, and $(0,1)$, respectively, and their dispersions are 1, $1/2$, and $1/2$. This implies (50)–(52). The zero-rate constraints in (48) are retained in every case.

At the curved/sum-rate interface $r = 0$, the curved weights equal $(1,1)$ and $V = 1$. At the curved/first-source interface $r = u$, they equal $(1,0)$ and $V = 1/2$; at the other one-source interface $r = u^{-1}$, they equal $(0,1)$ and $V = 1/2$. These values agree with the adjacent affine conditions. Lemma 3 specifies which strict one-source tuples are admissible boundary rate tuples. This completes the proof.

References

[1] C. E. Shannon, "A mathematical theory of communication," *Bell Syst. Tech. J.*, vol. 27, pp. 379–423, 623–656, Jul. and Oct. 1948.

[2] V. Strassen, "Asymptotische Abschätzungen in Shannons Informationstheorie," in *Trans. Third Prague Conf. Inf. Theory, Stat. Decision Functions, Random Processes*, J. Kožešník, Ed. Prague: Publishing House of the Czechoslovak Academy of Sciences, 1964, pp. 689–723, conference held at Liblice near Prague, June 5–13, 1962.

[3] I. Kontoyiannis and S. Verdú, "Optimal lossless data compression: Non-asymptotics and asymptotics," *IEEE Trans. Inf. Theory*, vol. 60, no. 2, pp. 777–795, Feb. 2014.

[4] C. E. Shannon, "Coding theorems for a discrete source with a fidelity criterion," in *IRE Int. Conv. Rec., Part 4*, vol. 7, 1959, pp. 142–163.

[5] A. Ingber and Y. Kochman, "The dispersion of lossy source coding," in *Proc. Data Compression Conf.*, Snowbird, UT, USA, Mar. 2011, pp. 53–62.

[6] V. Kostina and S. Verdú, "Fixed-length lossy compression in the finite blocklength regime," *IEEE Trans. Inf. Theory*, vol. 58, no. 6, pp. 3309–3338, Jun. 2012.

[7] K. B. Viswanatha, E. Akyol, and K. Rose, "The lossy common information of correlated sources," *IEEE Trans. Inf. Theory*, vol. 60, no. 6, pp. 3238–3253, Jun. 2014.

[8] P. Hassanzadeh, A. M. Tulino, J. Llorca, and E. Erkip, "Rate-memory trade-off for caching and delivery of correlated sources," *IEEE Trans. Inf. Theory*, vol. 66, no. 4, pp. 2219–2251, Apr. 2020.

[9] R. M. Gray and A. D. Wyner, "Source coding for a simple network," *Bell Syst. Tech. J.*, vol. 53, no. 9, pp. 1681–1721, Nov. 1974.

[10] S. Watanabe, "Second-order region for Gray–Wyner network," *IEEE Trans. Inf. Theory*, vol. 63, no. 2, pp. 1006–1018, Feb. 2017.

[11] L. Zhou, V. Y. F. Tan, and M. Motani, "Discrete lossy Gray–Wyner revisited: Second-order asymptotics, large and moderate deviations," *IEEE Trans. Inf. Theory*, vol. 63, no. 3, pp. 1766–1791, Mar. 2017.

[12] J. Liu, T. A. Courtade, P. Cuff, and S. Verdú, "Smoothing Brascamp–Lieb inequalities and strong converses of coding theorems," *IEEE Trans. Inf. Theory*, vol. 66, no. 2, pp. 704–721, Feb. 2020.

[13] L. Yu, "Gray–Wyner and mutual information regions for doubly symmetric binary sources and Gaussian sources," *IEEE Trans. Inf. Theory*, vol. 69, no. 10, pp. 6251–6268, Oct. 2023.

[14] I. Csiszár, "The method of types," *IEEE Trans. Inf. Theory*, vol. 44, no. 6, pp. 2505–2523, Oct. 1998.

[15] L. Zhou, V. Y. F. Tan, and M. Motani, "Second-order and moderate deviations asymptotics for successive refinement," *IEEE Trans. Inf. Theory*, vol. 63, no. 5, pp. 2896–2921, May 2017.

[16] V. Bentkus, "A Lyapunov-type bound in $\mathbb{R}^d$," *Theory Probab. Appl.*, vol. 49, no. 2, pp. 311–323, 2005.

[17] D. Cao, "Second-order image characterization with distortion and secrecy," arXiv:2610.09472v1, Oct. 2026.

[18] V. Y. F. Tan and O. Kosut, "On the dispersions of three network information theory problems," *IEEE Trans. Inf. Theory*, vol. 60, no. 2, pp. 881–903, Feb. 2014.

[19] D. Slepian and J. K. Wolf, "Noiseless coding of correlated information sources," *IEEE Trans. Inf. Theory*, vol. 19, no. 4, pp. 471–480, Jul. 1973.

[20] R. Nomura and T. S. Han, "Second-order Slepian–Wolf coding theorems for non-mixed and mixed sources," *IEEE Trans. Inf. Theory*, vol. 60, no. 9, pp. 5553–5572, Sep. 2014.

[21] A. No, A. Ingber, and T. Weissman, "Strong successive refinability and rate-distortion-complexity tradeoff," *IEEE Trans. Inf. Theory*, vol. 62, no. 6, pp. 3618–3635, Jun. 2016.

[22] B. Rimoldi, "Successive refinement of information: characterization of the achievable rates," *IEEE Trans. Inf. Theory*, vol. 40, no. 1, pp. 253–259, Jan. 1994.

[23] A. D. Wyner and J. Ziv, "The rate-distortion function for source coding with side information at the decoder," *IEEE Trans. Inf. Theory*, vol. 22, no. 1, pp. 1–10, Jan. 1976.

[24] S. Watanabe, S. Kuzuoka, and V. Y. F. Tan, "Nonasymptotic and second-order achievability bounds for coding with side-information," *IEEE Trans. Inf. Theory*, vol. 61, no. 4, pp. 1574–1605, Apr. 2015.

[25] X. Li and C. T. Li, "New second-order achievability bounds for coding with side information via type deviation convergence," in *Proc. IEEE Int. Symp. Inf. Theory*, Guangzhou, China, Jun. 2026, pp. 1–6.

[26] L. Zhou, "Exact second-order asymptotics for the discrete Wyner–Ziv problem," Oct. 2026, arXiv:2610.11998, version 1.

[27] L. Zhou and M. Motani, "Finite blocklength lossy source coding for discrete memoryless sources," *Found. Trends Commun. Inf. Theory*, vol. 20, no. 3, pp. 157–389, Jun. 2023.

[28] A. D. Wyner, "The common information of two dependent random variables," *IEEE Trans. Inf. Theory*, vol. 21, no. 2, pp. 163–179, Mar. 1975.

[29] E. Sula and M. Gastpar, "The Gray–Wyner network and Wyner's common information for Gaussian sources," *IEEE Trans. Inf. Theory*, vol. 68, no. 2, pp. 1369–1384, Feb. 2022.

[30] T. M. Cover and J. A. Thomas, *Elements of Information Theory*, 2nd ed. Hoboken, NJ, USA: Wiley-Interscience, 2006.

[31] R. C. Yavas, V. Kostina, and M. Effros, "Third-order analysis of channel coding in the small-to-moderate deviations regime," *IEEE Trans. Inf. Theory*, vol. 70, no. 9, pp. 6139–6170, Sep. 2024.

[32] A. D. Wyner, "The rate-distortion function for source coding with side information at the decoder–II: General sources," *Inf. Control*, vol. 38, no. 1, pp. 60–80, Jul. 1978.

[33] C. Heegard and T. Berger, "Rate distortion when side information may be absent," *IEEE Trans. Inf. Theory*, vol. 31, no. 6, pp. 727–734, Nov. 1985.